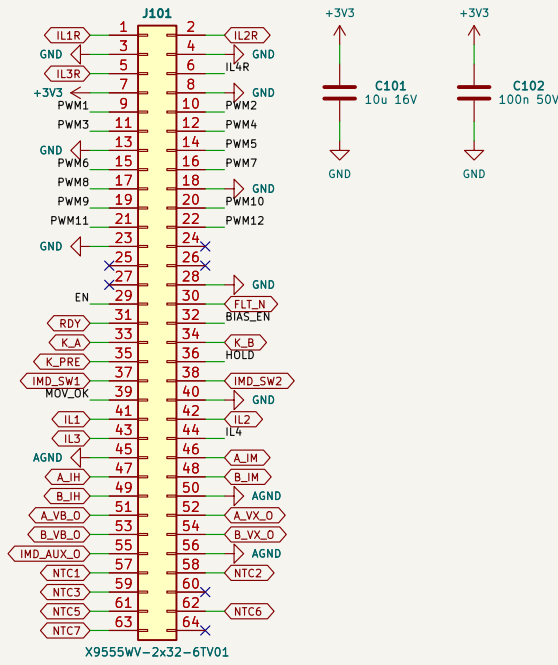


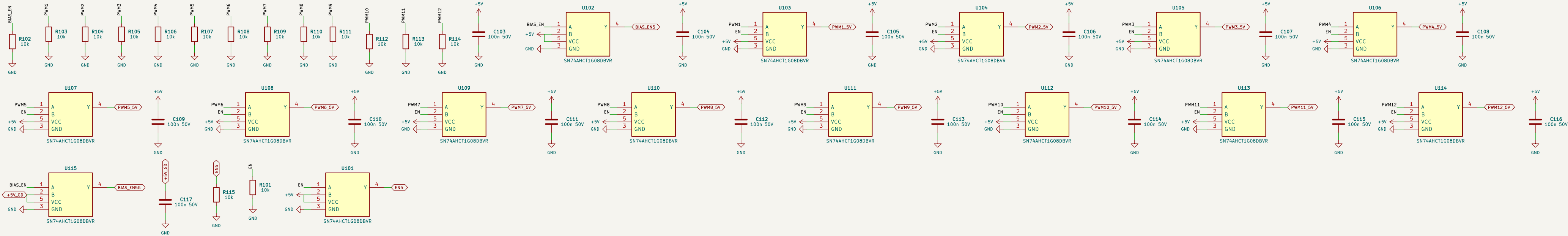
	1	2	3	4	5	6	7	8
A	PV–PWR power board Rev A2 2026–10–05 Schematics authored by Chinmoy Bhuyan PV–P75: 3 interleaved FSBB phases, gate drive, sensing, ports, aux (D–044)							
B	01 PC connector, command buffers, status interfaces.PC (2x32) to the control board, 3.3 -> 5 V command buffers, RDY / FLT_N, +5 V supervisor, port status glue, BUS– star points File: sheets/01_pc.kicad_sch	02 Phase 1 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/02_ph1_power.kicad_sch	03 Phase 1 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/03_ph1_gda.kicad_sch	04 Phase 1 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/04_ph1_gdb.kicad_sch				
C	05 Phase 2 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/05_ph2_power.kicad_sch	06 Phase 2 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/06_ph2_gda.kicad_sch	07 Phase 2 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/07_ph2_gdb.kicad_sch	08 Phase 3 power stage, bias, sensing Legs A/B (2 x SG2M040170HJ per switch), decoupling, dampers, bank share, inductor + STK–HQ/A 75, NTC inputs, gate bias (gdrv.bias_phase) File: sheets/08_ph3_power.kicad_sch				
D	09 Phase 3 leg A gate drive (S1, S2) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/09_ph3_gda.kicad_sch	10 Phase 3 leg B gate drive (S3, S4) gdrv.channel() rev 7, bias='ext': NSI6651ASC–Q1 +18/–3.5 V, 3.75/2.5 ohm per device, Miller clamps, SC booster 22R, interlock + dead–time stretch, DESAT to the drain File: sheets/10_ph3_gdb.kicad_sch	11 Port A power path (pv) Varistor Y + monitor File: sheets/11_leanA_power.kicad_sch	12 Port A sensing and interlocks References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks File: sheets/12_leanA_ctrl.kicad_sch				
E	13 Port A coil drives Contactor: interlock logic, fail–safe high side, economiser; passive bank bleeder File: sheets/13_leanA_drive.kicad_sch	13 Port B power path (battery) Varistor Y + monitor, aR fuses File: sheets/13_leanB_power.kicad_sch	14 Port B sensing and interlocks References, BUS– shunt amplifier + hold–off / OC windows, dividers, dV amplifier, polarity and dV interlocks File: sheets/14_leanB_ctrl.kicad_sch	15 Port B coil drives Contactor + precharge relay: interlock logic, fail–safe high side, economiser; passive bank bleeder File: sheets/15_leanB_drive.kicad_sch				
F	15 Terminals, EMI filter, insulation test CAUTION! no one should ever use the EMI filter or the insulation test on each build (output 24 V / PV–PWR design check k6VGD41MB38920) rev 3 from 562436 from is 8x.GD ramp–up and discharge), +5V_GD supervisor, SELV connector to the control board (MD crossing), SELV (AUX04T1 from force where (AUX–HSELV), CC output 000). Shared heatsink: Tj 107 C at 45 C inlet (R_sa target); heatsink OT trip must be <= 90 C. Dead time 211–554 ns. Sensor delay 0.33 us, error 1.8 % at 45 A after calibration. Live 24 V 19.8 W (economiser) <= aux live rating 20.0 W; module aux 60 W. Full fan speed is only guaranteed while the converter is switching. File: sheets/15_terminals_imd.kicad_sch	16 Live 5 V / 3.3 V, SELV connector Live 23.62–24.68 V (FB); SELV 20.3–29.8 V over the load table, gates off 20.3–28.2 V; hold–up 52 ms at 11 W; status = TPS3700 OUTA (UV); OUTB (OV) pulls SS low File: sheets/16_supplies.kicad_sch	17 AUX75 input taps, OR, film One tap per port (+ pole, terminal side), return = BUS–; 2 x 20R R pulse–rated ahead of the 1 A / 1000 VDC fuse, BYG10Y OR; 350 V (AUX04T1 from force where (AUX–HSELV), CC output 000). File: sheets/17_aux75_input.kicad_sch	18 AUX75 start–up, brown–in, OV, timer As AUX–HV rev C1 sheet 02: 3 x BSS126 cascode, TPS3710 brown–in 206–215 V, TLV3202 OV lockout 1107–1162 V and overload timer 35–80 ms File: sheets/18_aux75_startup.kicad_sch	Not bench–validated. Values read from sim/out/*.json; checks: design_check() Sheet 1 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS–; gate islands functional; SELV winding reinforced; HV–PE basic Sheet: / File: PV–PWR.kicad_sch Title: PV–PWR power board – Cover / Hierarchy Size: A3 Date: 2026–10–05 Rev: A2 KiCad E.D.A. 10.0.6 Id: 1/23			

PC CONNECTOR (INTERFACES.PC), LIVE



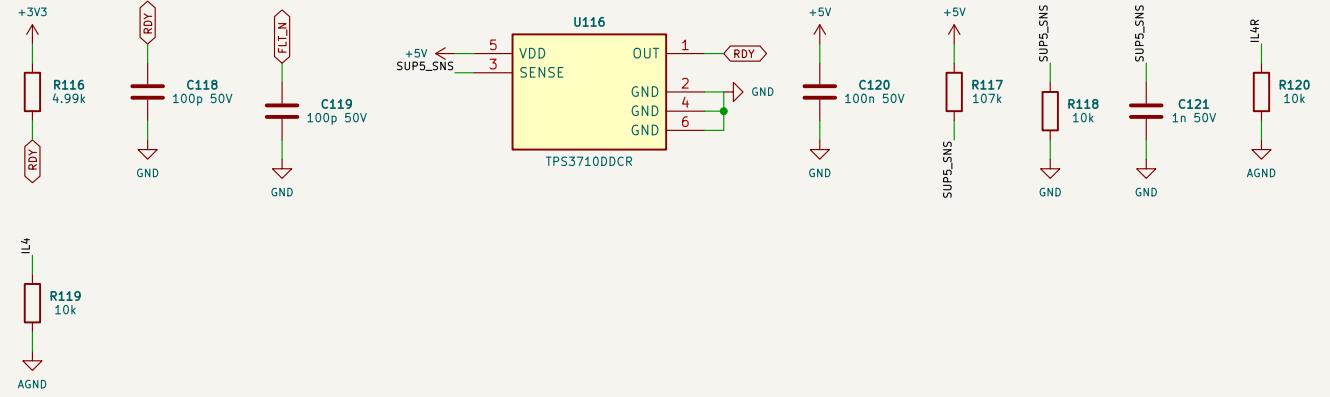
Both boards live: GND = AGND = BUS-; +3V3 goes TO the control board (rev A2; the former +24V / +5V pins carry the sensor references IL1R-IL4R). Commands are 3.3 V logic after the control board's trip latch; every command input is pulled low here (open = off).
PV-P75: PWM13-16, NTC4 and NTC6 not connected; IL4 and IL4R pulled to AGND (0 V = no phase 4).

COMMAND INPUTS: PULL-DOWNS AND 3.3 -> 5 V BUFFERS (GDRV.INPUT_BUFFERS)



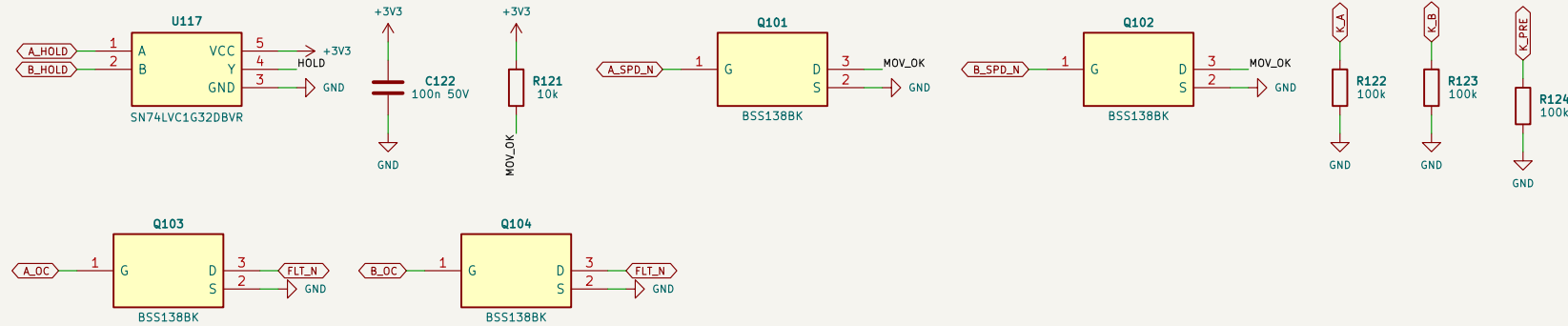
NSI6651 inputs are specified at VCC1 = 5 V only (VINH <= 3.5 V, p7) and the SN6505B EN needs 0.7 VCC: one SN74AHCT1G08 (VIN 2.0 V) per line. Each PWM is ANDed with EN here as well, so a low EN removes every command within 8 ns, ahead of the drivers' RST/EN filter (0.48-0.8 us).

RDY (WIRED-AND) AND FLT_N (WIRED-OR)



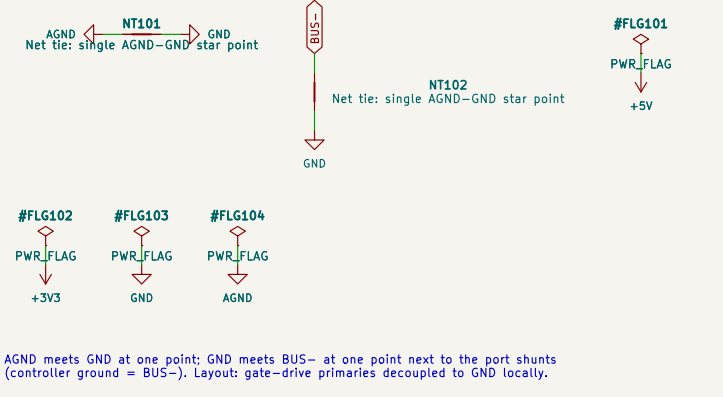
RDY: every NSI6651 RDY (VCC1 and VCC2 UVLO), the +5V and +5V_GD supervisors and the aux status (live 24 V UV), pulled up HERE to +3V3 (4.99k). FLT_N: every NSI6651 /FLT + port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
+5V supervisor TPS3710 107k/10k; RDY low below 4.52-4.69 V (AHC2 VCC <= 4.5 V, bias headroom).

PORT STATUS TO PC: HOLD, MOV_OK, PORT OVER-CURRENT INTO FLT_N



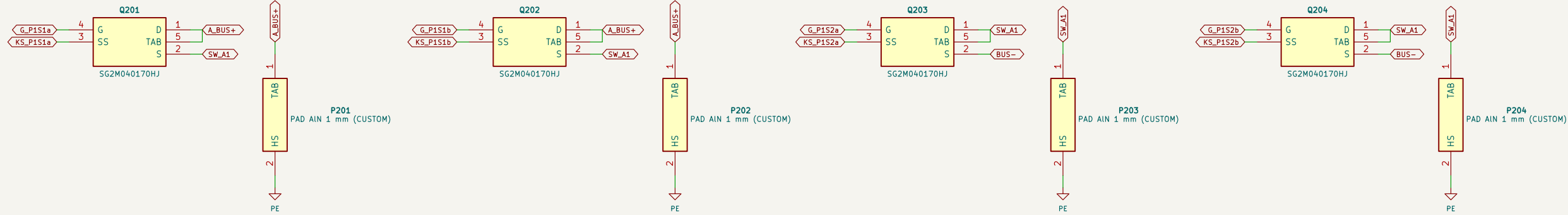
HOLD = A_HOLD OR B_HOLD ([Lport] >= 0.97-1.03 kA, contactor held), MOV_OK high only while BOTH varistor monitor loops conduct (each reads 'open' below -200 V on its port). Port OC ([1] >= 387-413 A, lean_shunt) pulls FLT_N low (open drain); the control board's latch trips without firmware (4R4-fuse gap, port-4prc).
K_A / K_B / K_PRE pulled low: open = coil off (the hardware interlocks follow on the port sheets).

GROUND STAR POINTS



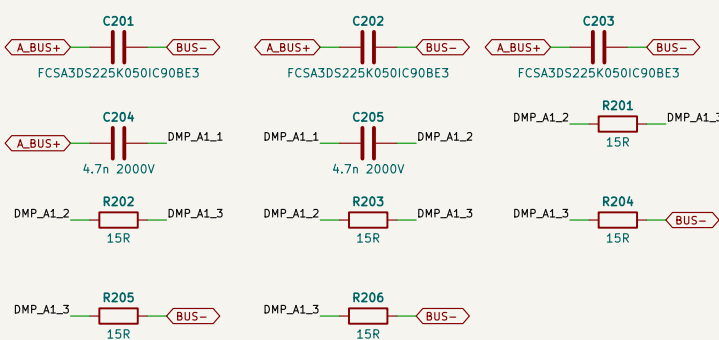
AGND meets GND at one point; GND meets BUS- at one point next to the port shunts (controller ground = BUS-); Layout: gate-drive primaries decoupled to GND locally.

PHASE 1 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



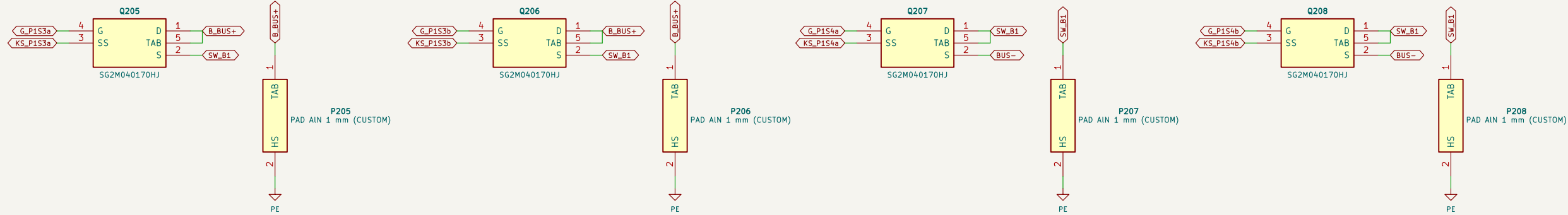
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG A DECOUPLING + RC DAMPER



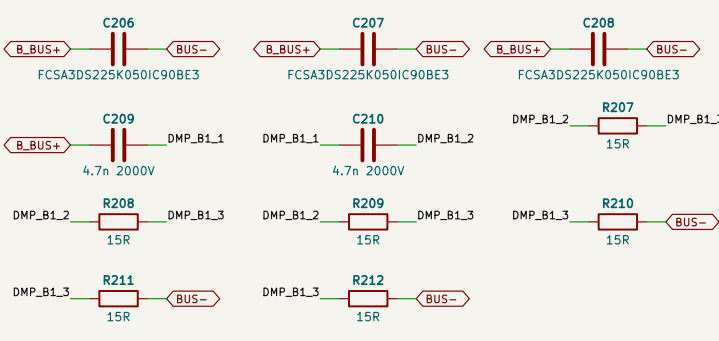
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 1 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



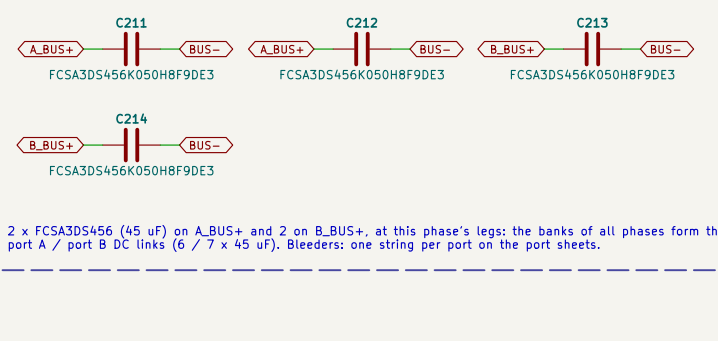
Kevin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 1 LEG B DECOUPLING + RC DAMPER



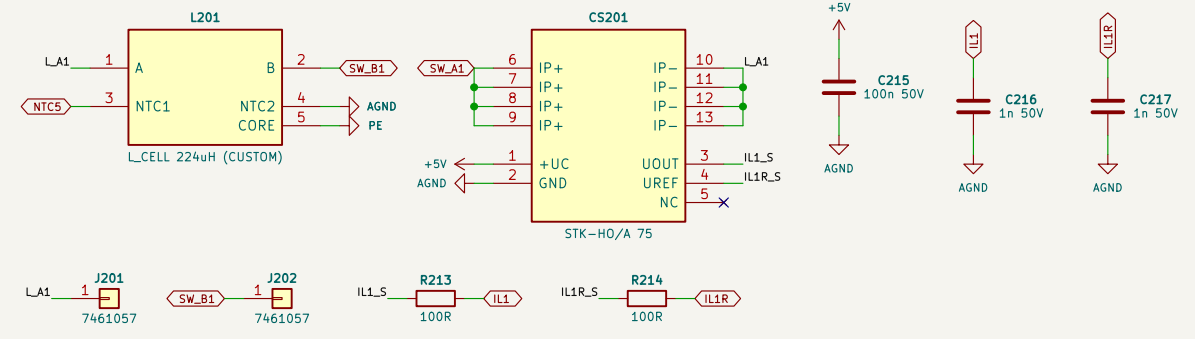
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 1 SHARE OF THE PORT BANKS



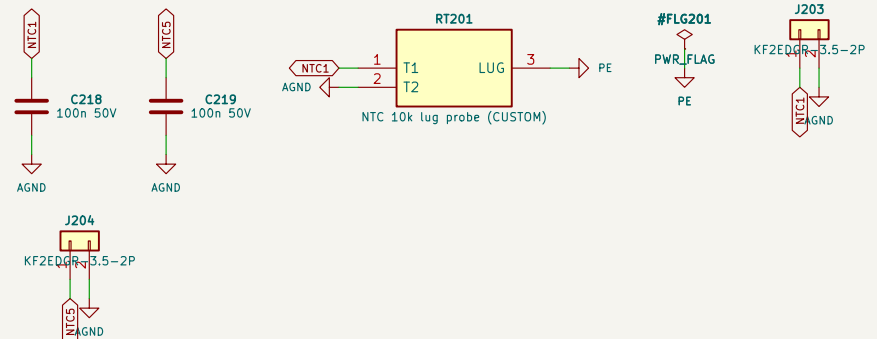
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 1 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



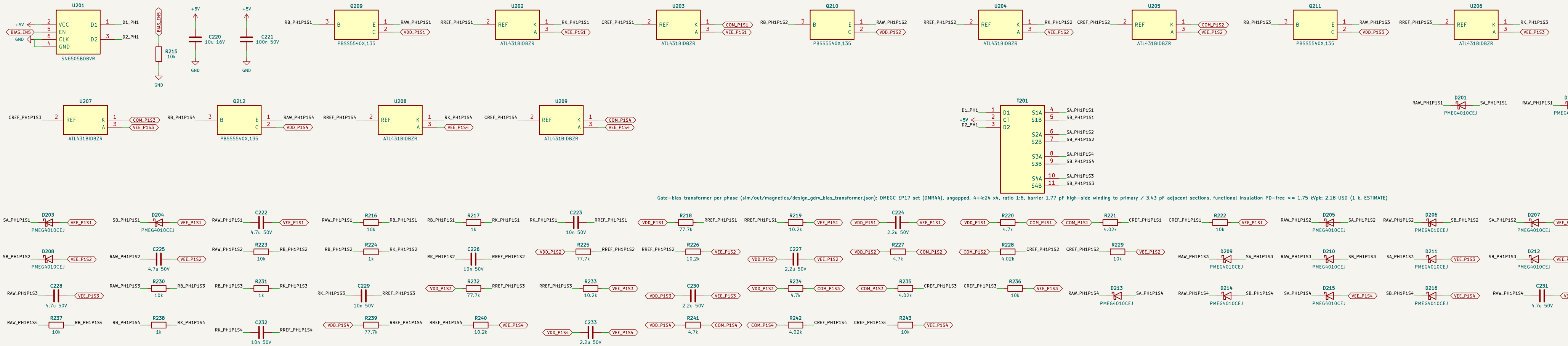
Lead A from SW.A1 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; += SW.A -> SW.B) to stud
LA1: lead B on stud SW.B1. IL1 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A;
100R = 1 nF at the source, IL1R = the sensor's own (ref pin 6), 100R + 1 nF (PVR-CTL references its trip
window to IL1. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 1 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



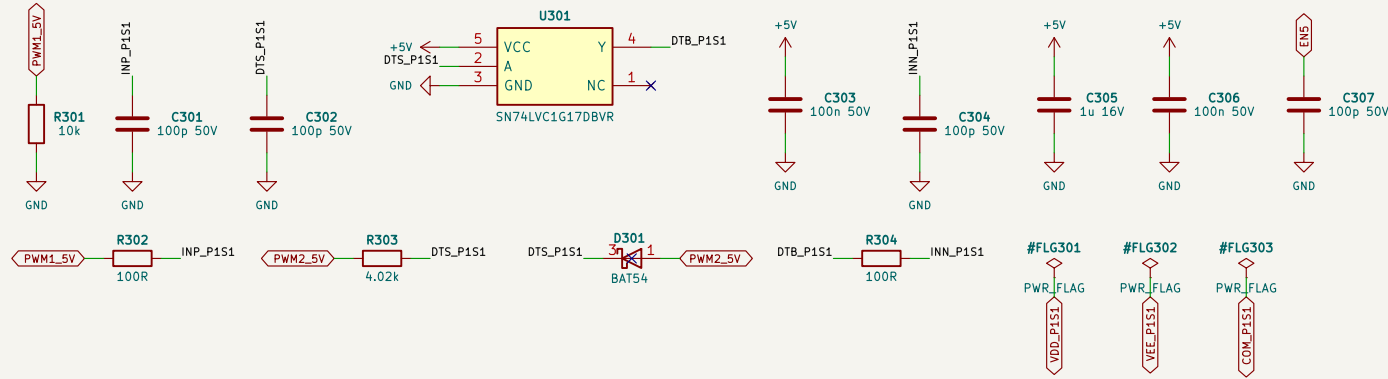
NTC1 = heatsink section of phase 1: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC5 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH1: GATE BIAS (PER PHASE)



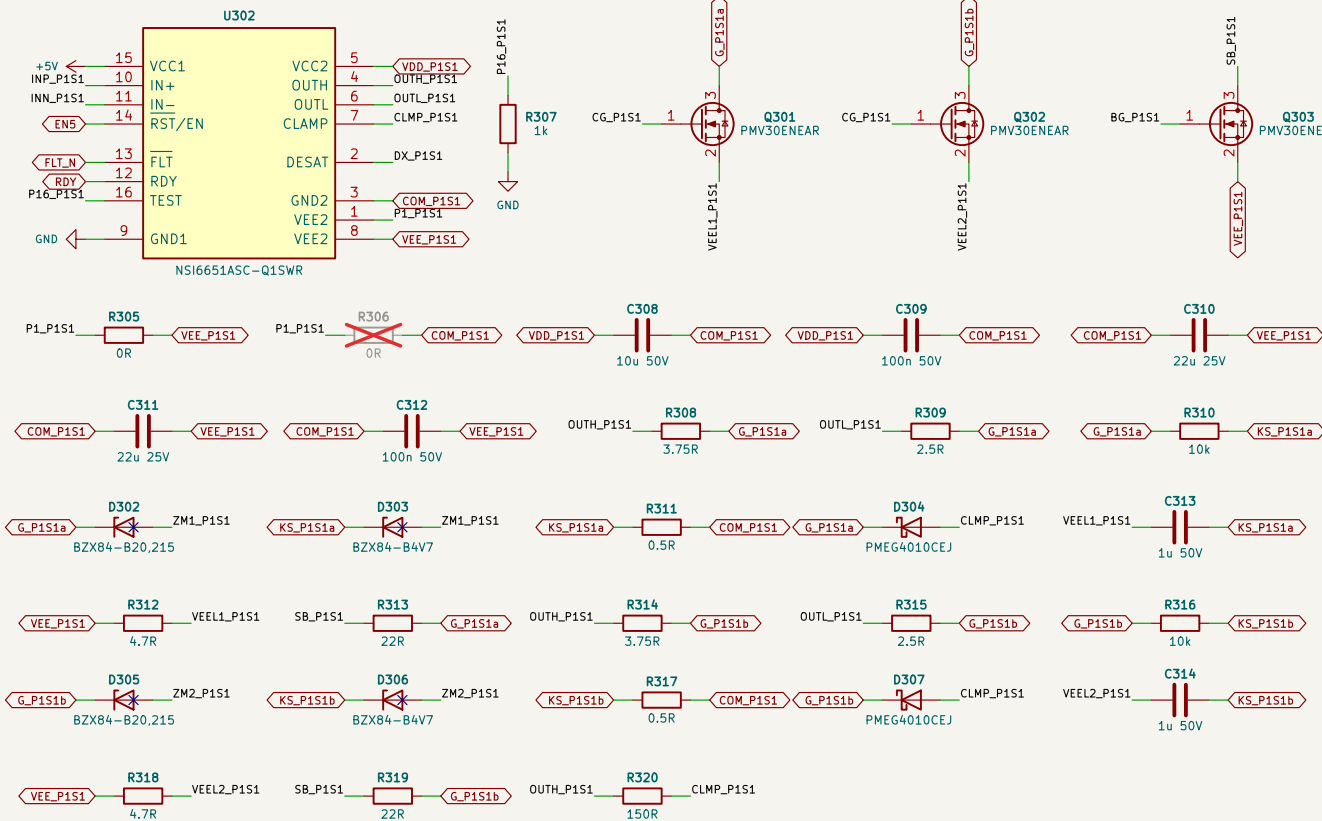
SN6505B push-pull 420 kHz from +5V, BIAS_EN high = on; T.BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P1S1: LOGIC SIDE



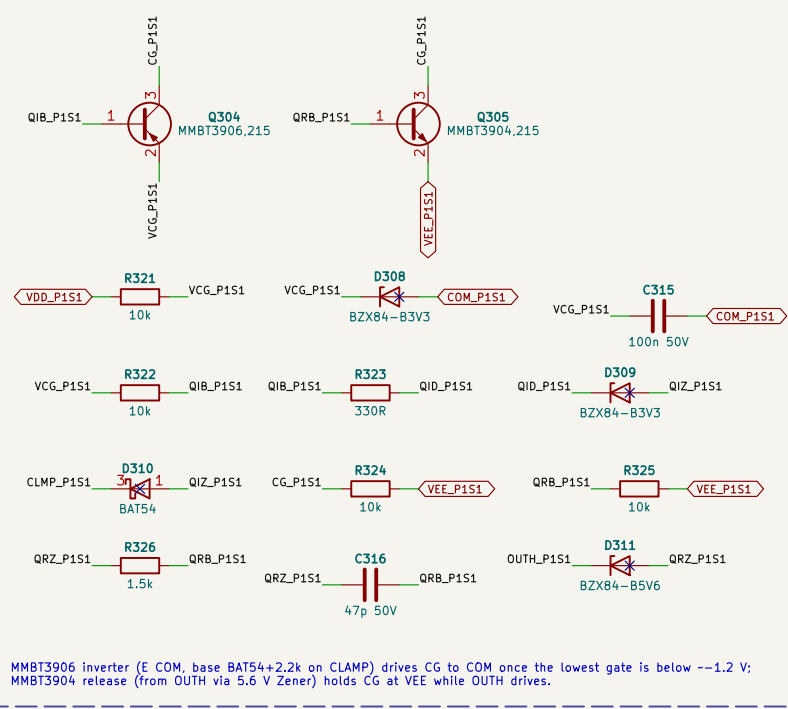
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S1: DRIVER AND 2 GATES



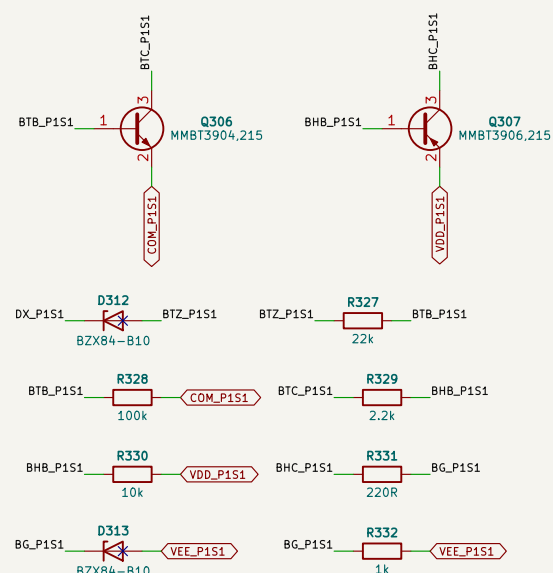
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S1: MILLER CLAMP CONTROL



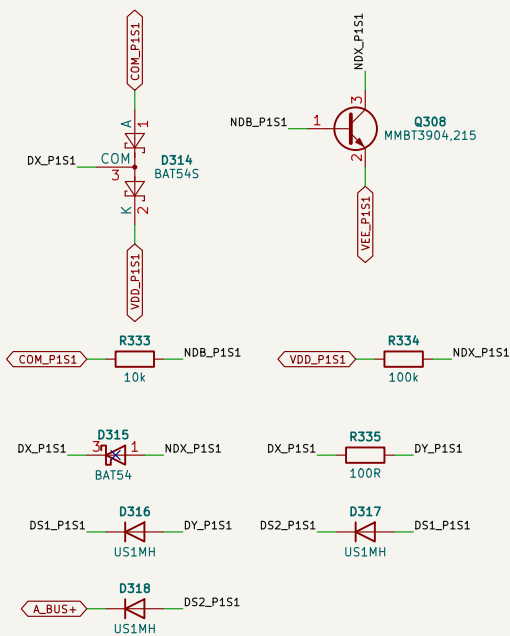
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S1: SHORT-CIRCUIT BOOSTER



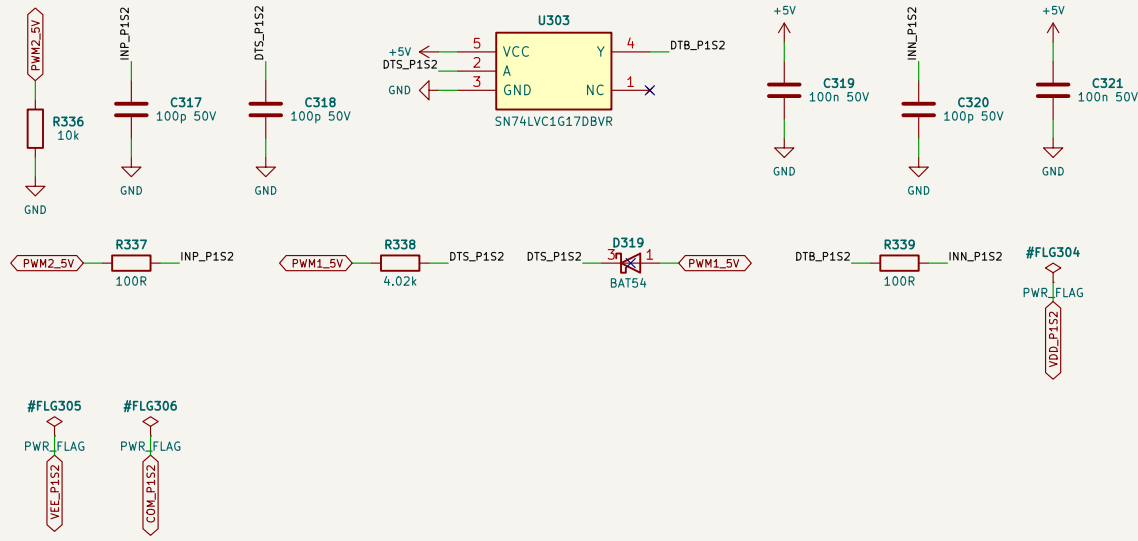
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P1S1: DESAT



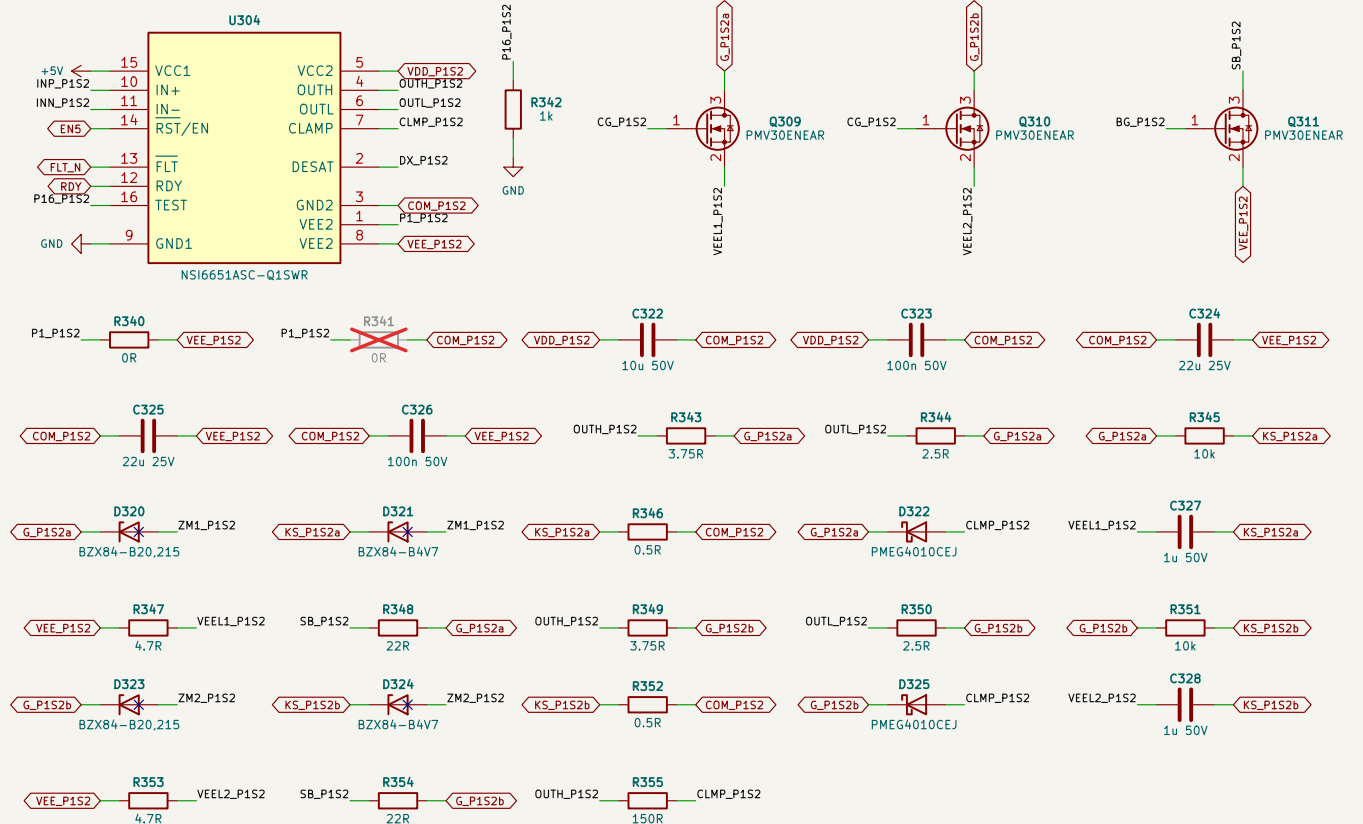
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdb footprint DNP.

P1S2: LOGIC SIDE



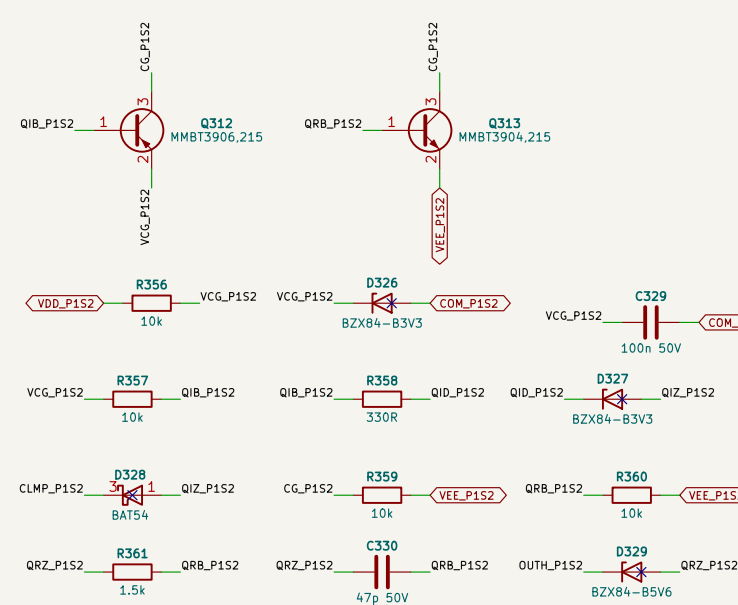
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P1S2: DRIVER AND 2 GATES



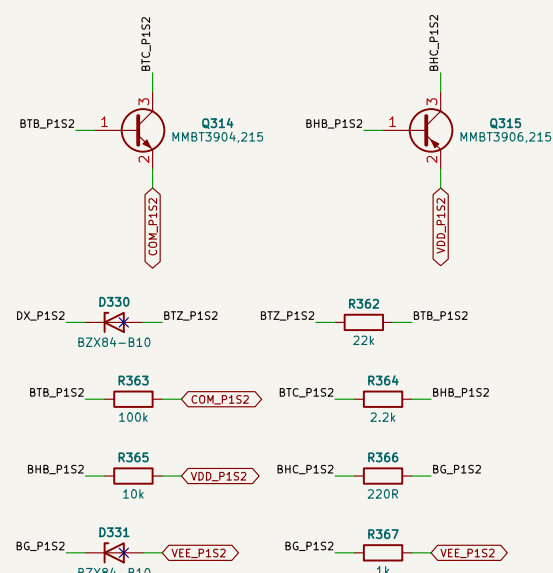
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P1S2: MILLER CLAMP CONTROL



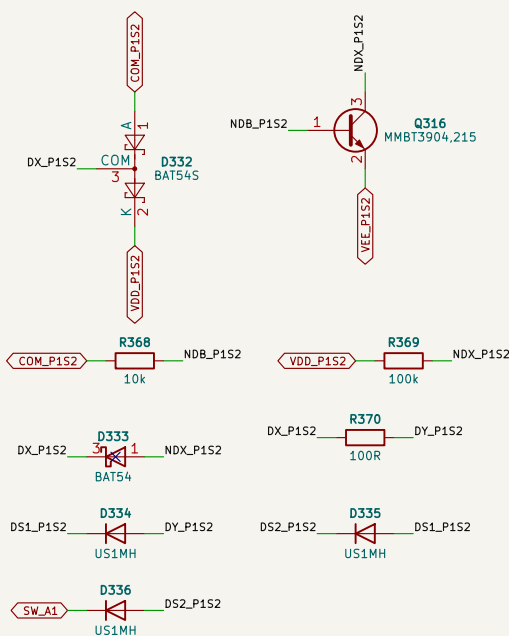
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P1S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V; per device AQ3400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P1S2: DESAT

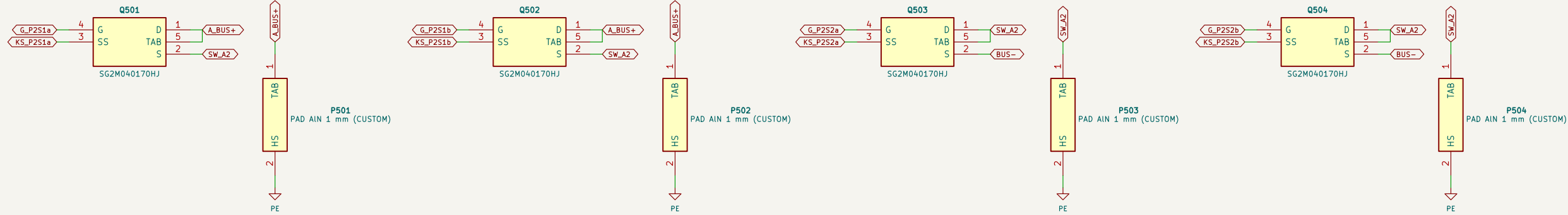


DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain, BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdb footprint DNP.

LEG A: LAYOUT RULES

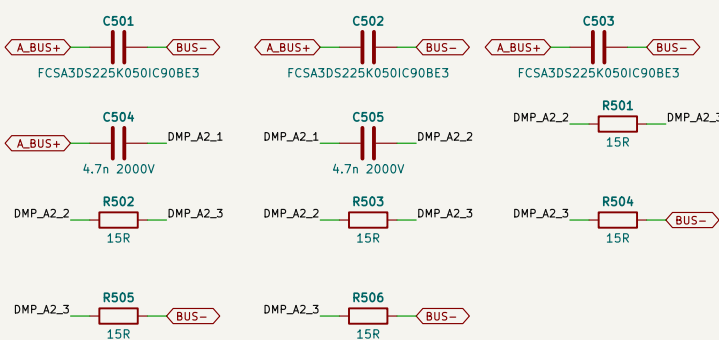
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PDI (IC-22). Miller hold (topside, gnd): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PHASE 2 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



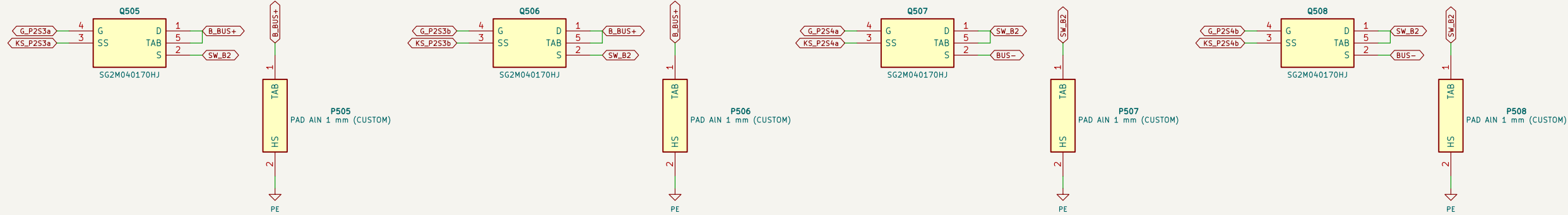
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG A DECOUPLING + RC DAMPER



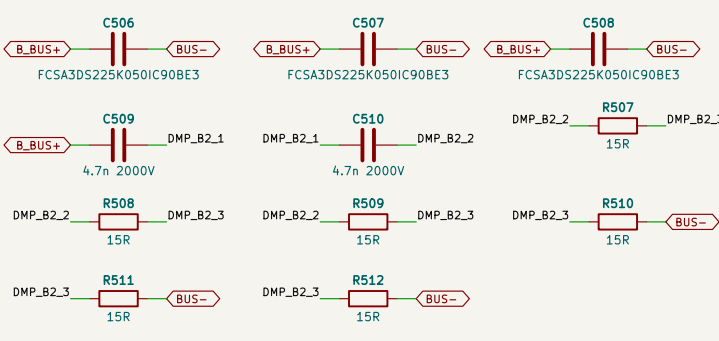
3 x FCSA3DS225 at the S1/S2 pins; commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV CGO in series; loop at the pins <= 3 nH.

PHASE 2 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



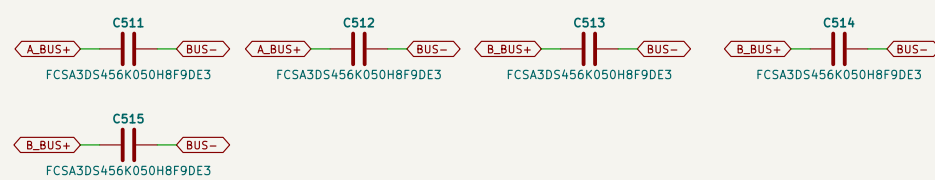
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own AIN ceramic (aluminum nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 2 LEG B DECOUPLING + RC DAMPER



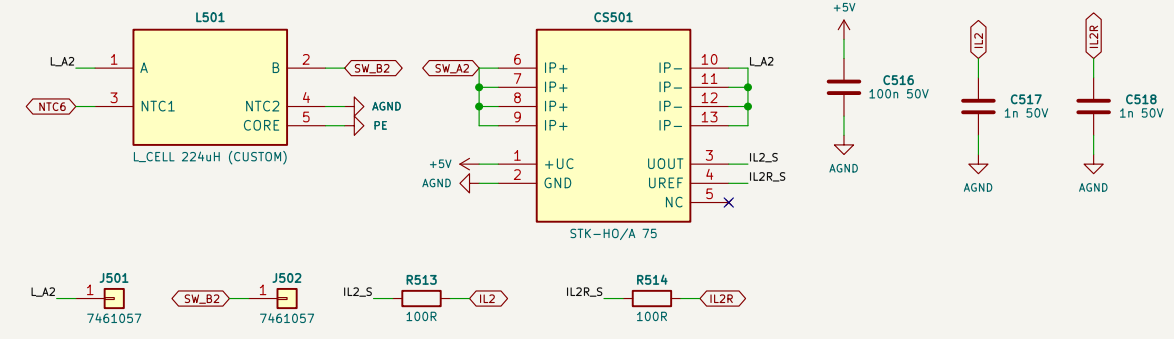
3 x FCSA3DS225 at the S3/S4 pins; commutation loop <= 20 nH; bank <= 15 nH from the leg (LAYOUT, leg deck). Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV CGO in series; loop at the pins <= 3 nH.

PHASE 2 SHARE OF THE PORT BANKS



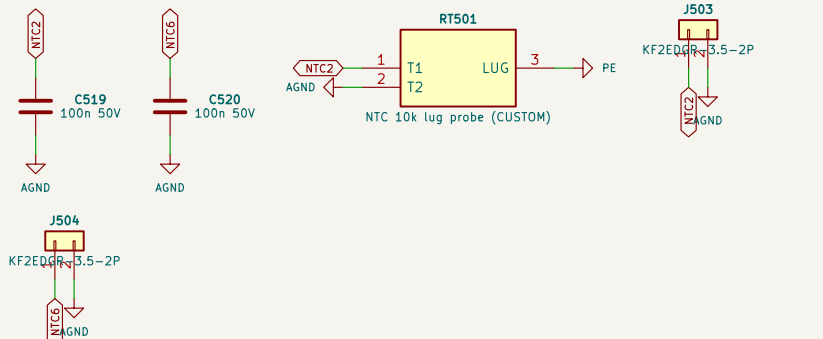
2 x FCSA3DS456 (45 uF) on A.BUS+ and 3 on B.BUS+, at this phase's legs; the banks of all phases form the port A / port B DC links (6 / 7 x 45 uF); the third on B.BUS+ is the battery side's seventh (D-056; full-power load rejection). Bleeders: one string per port on the port sheets.

PHASE 2 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



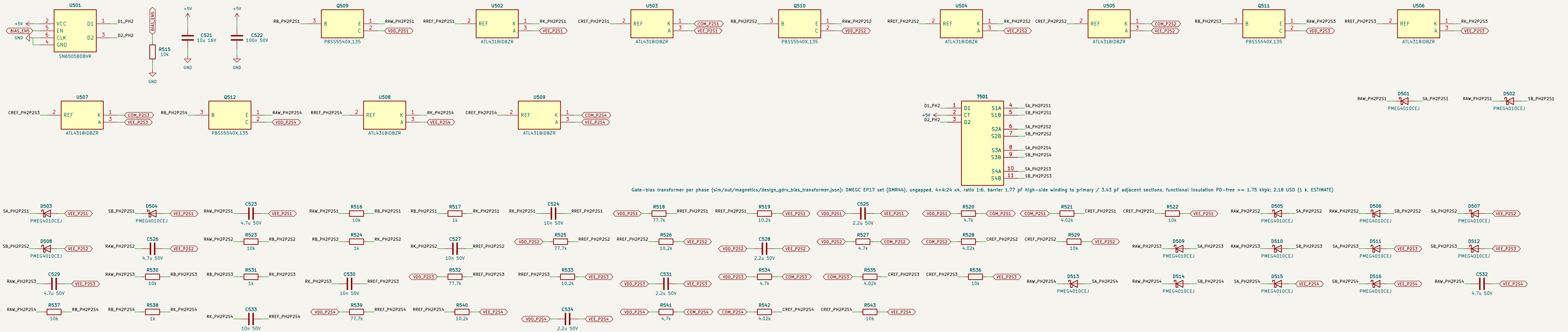
Lead A from SW_A2 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out: + = SW_A -> SW_B) to stud LA2; lead B on stud SW_B2. IL2 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-4.5 V over +/-187.5 A; 100R = 1 nF at the source; IL2R = the sensor's own Uref (pin 4), 100R = 1 nF; PV-CTL references its trip window to it. Primary at the switch node; functional barrier (4 kV rms, 8 kV impulse).

PHASE 2 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC2 = heatsink section of phase 2; lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10). NTC6 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

PH2: GATE BIAS (PER PHASE)



SN6505B push-pull 420 kHz from +5V. BIAS_EN high = on; T_BIAS4 custom; functional insulation (PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431, COM split.

[illegible]

The schematic diagram illustrates the internal circuitry of the NS6651ASC-Q15WR, a power MOSFET driver. The central component is the U602 MOSFET driver IC, which is configured as a common source amplifier. Its VCC1 pin is connected to a 5V supply, and its VCC2 pin is connected to a 15V supply. The U602's IN+ pin is connected to the 15V supply, and its IN- pin is connected to the 5V supply. The U602's OUTL pin is connected to the gate of the P1_P251 MOSFET. The P1_P251 MOSFET is a power MOSFET that drives the gate of the Q601 PMV30E01 MOSFET. The Q601 is a power MOSFET that drives the load. The circuit is powered by a 5V supply and a 15V supply. The U602 is connected to the 5V supply via its VCC1 pin and to the 15V supply via its VCC2 pin. The P1_P251 is connected to the 15V supply via its VDD_P251 pin and to ground via its VEE_P251 pin. The Q601 is connected to the 15V supply via its VDD_P251 pin and to ground via its VEE_P251 pin. The circuit includes various resistors and capacitors for timing and biasing.

Figure 1 shows two schematic diagrams of the 100nF 50V electrolytic capacitor (C614) in the Q604 and Q605 MMBT3906.215 transistors. The diagrams illustrate the internal structure of the capacitor, including the VCC_P251, VEE_P251, and COM_P251 connections, and the 100nF 50V electrolytic capacitor (C614) connected to the VEE_P251 and COM_P251 pins.

Q604 MMBT3906.215: The diagram shows the VCC_P251 pin connected to the VEE_P251 pin through a 10k resistor (R621). The VEE_P251 pin is connected to the COM_P251 pin through a 100nF 50V electrolytic capacitor (C614). The COM_P251 pin is connected to the VEE_P251 pin through a 100nF 50V electrolytic capacitor (C614).

Q605 MMBT3906.215: The diagram shows the VCC_P251 pin connected to the VEE_P251 pin through a 10k resistor (R622). The VEE_P251 pin is connected to the COM_P251 pin through a 100nF 50V electrolytic capacitor (C614). The COM_P251 pin is connected to the VEE_P251 pin through a 100nF 50V electrolytic capacitor (C614).

[illegible]

[illegible]

.....

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[illegible]

Figure 1 displays the schematic diagrams for the 12 channels of the Q1000, labeled Q1B_P252 through Q12B_P252. Each channel schematic shows the internal components and connections:

- Q1B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q1B_P252 and its emitter to VEE_P252.
- Q2B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q2B_P252 and its emitter to VEE_P252.
- Q3B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q3B_P252 and its emitter to VEE_P252.
- Q4B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q4B_P252 and its emitter to VEE_P252.
- Q5B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q5B_P252 and its emitter to VEE_P252.
- Q6B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q6B_P252 and its emitter to VEE_P252.
- Q7B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q7B_P252 and its emitter to VEE_P252.
- Q8B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q8B_P252 and its emitter to VEE_P252.
- Q9B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q9B_P252 and its emitter to VEE_P252.
- Q10B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q10B_P252 and its emitter to VEE_P252.
- Q11B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q11B_P252 and its emitter to VEE_P252.
- Q12B_P252:** Features a 10k resistor connected to VDD_P252, a 100nF capacitor connected to VCG_P252, and a 2N3904 transistor with its base connected to Q12B_P252 and its emitter to VEE_P252.

[illegible]

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string <= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star: Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C_{VL} at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string <= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (ngspice, gdrv): die +1.41 V vs VGS(th) min 1.94 V at 175 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

P253: LOGIC SIDE

The schematic shows the logic side of the P253 component. The central component is the SN74LVC1G170BVR (U701), a 1.8V CMOS inverter. Its pins are connected as follows:

- VCC (Pin 5):** Connected to +5V.
- A (Pin 2):** Connected to DTS_P253.
- GND (Pin 3):** Connected to GND.
- Y (Pin 4):** Connected to DTB_P253.
- NC (Pin 1):** Connected to GND.

The input signal DTS_P253 is connected to the inverter's input (A) and also to the input of the D701 (BAT54) diode. The output of the inverter (Y) is connected to the output of the D701. The output of the D701 is connected to the input of the R704 (100R) resistor, which is connected to the input of the INN_P253 pin. The output of the INN_P253 pin is connected to the input of the C703 (100n 50V) capacitor, which is connected to GND. The output of the C703 is connected to the input of the C704 (100p 50V) capacitor, which is connected to GND. The output of the C704 is connected to the input of the C705 (1u 16V) capacitor, which is connected to GND. The output of the C705 is connected to the input of the C706 (100n 50V) capacitor, which is connected to GND. The output of the C706 is connected to the input of the #FLG701 and #FLG702 pins. The output of the #FLG701 pin is connected to the input of the PWR_FLAG pin. The output of the #FLG702 pin is connected to the input of the PWR_FLAG pin. The output of the PWR_FLAG pin is connected to the input of the COM_P253 pin.

Legend:

- IN+ = own PWM
- IN- = complementary switch, falling edge stretched 4.02k/100p
- (interlock + min dead time = max(firmware, hardware)).
- Bias PG (active low) pulls RDY low.

U702

+5V < 1.5	VCC1	VCC2	VDD_P253
INP_P253 1.0	IN+	OUTH	OUTH_P253
INM_P253 1.1	IN-	OUTL	OUTL_P253
ENS 1.4	RST/EN	CLAMP	CLMP_P253
FLT_N 1.3	FLT	DESAT	DX_P253
RDT 1.2	RDY		
P1_P253 1.6	GND2	TEST	COM_P253
	GND1	VEE2	VEE_P253
		VEE2	VEE_P253

NS16651ASC-Q1SWR

P1_P253 R705 VEE_P253 OR

P1_P253 R706 COM_P253 OR

VDD_P253 C707 10u 50V COM_P253

VDD_P253 C708 100n 50V COM_P253

COM_P253 C709 22u 25V VEE_P253

COM_P253 C710 22u 25V VEE_P253

COM_P253 C711 100n 50V VEE_P253

OUTH_P253 R708 3.75R G_P253a

OUTL_P253 R709 2.5R G_P253a

G_P253a R710 10k KS_P253a

G_P253a D702 ZML_P253 BZX84-B20.15 KS_P253a D703 ZML_P253 BZX84-B4V7 KS_P253a R711 0.5R COM_P253 G_P253a D704 CLMP_P253 PMEGA010CEJ VEE1_P253 C712 1u 50V KS_P253a

VEE_P253 R712 4.7R VEE1_P253 SB_P253 R713 22R G_P253a

OUTH_P253 R714 3.75R G_P253b

OUTL_P253 R715 2.5R G_P253b

G_P253b R716 10k KS_P253b

G_P253b D705 ZML_P253 BZX84-B20.15 KS_P253b D706 ZML_P253 BZX84-B4V7 KS_P253b R717 0.5R COM_P253 G_P253b D707 CLMP_P253 PMEGA010CEJ VEE2_P253 C713 1u 50V KS_P253b

VEE_P253 R718 4.7R VEE1_P253 SB_P253 R719 22R G_P253a

OUTH_P253 R720 150R CLMP_P253

Per device: Ron/Off: 0.5R Kelvin R to a Zener (about +21 V / -5.7 V) on its own Kelvin net.
 Miller hold: 10-MOSFET clamp (PMV30E01A) gate-pin to a local VEE (5u on the Kelvin pin). LAYOUT: FEI + 1u at the device, loop <= 1 nH.

[illegible]

P2S3: SHORT-CIRCUIT BOOSTER

The schematic diagram illustrates the P2S3 Short-Circuit Booster circuit. It features two main sections, one for the positive supply (P2S3) and one for the negative supply (P2S4). Each section includes a MOSFET (Q706 and Q707), a gate driver (R727, R728, R729, R730, R731), and a current sense resistor (R712, R713). The MOSFETs are connected to the positive and negative supply rails, respectively. The gate drivers are connected to the gates of the MOSFETs. The current sense resistors are connected in series with the MOSFETs. The circuit is designed to provide a short-circuit current boost to the P2S3 and P2S4 rails.

Positive Supply Section (P2S3):

- Q706:** MOSFET connected to P2S3.
- R727:** 22k resistor connected to BTZ_P2S3 and BTB_P2S3.
- R728:** 100k resistor connected to BTB_P2S3 and COM_P2S3.
- R729:** 2.2k resistor connected to BTC_P2S3 and BHB_P2S3.
- R730:** 10k resistor connected to BHB_P2S3 and VDD_P2S3.
- R731:** 220R resistor connected to BHC_P2S3 and BG_P2S3.
- R712:** Current sense resistor connected to BTZ_P2S3 and BZX84-810.

Negative Supply Section (P2S4):

- Q707:** MOSFET connected to P2S4.
- R727:** 22k resistor connected to BTZ_P2S3 and BTB_P2S3.
- R728:** 100k resistor connected to BTB_P2S3 and COM_P2S3.
- R729:** 2.2k resistor connected to BTC_P2S3 and BHB_P2S3.
- R730:** 10k resistor connected to BHB_P2S3 and VDD_P2S3.
- R731:** 220R resistor connected to BHC_P2S3 and BG_P2S3.
- R713:** Current sense resistor connected to BTZ_P2S3 and BZX84-810.

Common Connections:

- COM_P2S3:** Connected to the positive supply rail.
- VDD_P2S3:** Connected to the negative supply rail.
- VEE_P2S3:** Connected to the negative supply rail.

Component Values:

- R727:** 22k
- R728:** 100k
- R729:** 2.2k
- R730:** 10k
- R731:** 220R
- R712:** Current sense resistor
- R713:** Current sense resistor

Notes:

- The circuit is designed to provide a short-circuit current boost to the P2S3 and P2S4 rails.
- The MOSFETs (Q706 and Q707) are connected to the positive and negative supply rails, respectively.
- The gate drivers (R727, R728, R729, R730, R731) are connected to the gates of the MOSFETs.
- The current sense resistors (R712, R713) are connected in series with the MOSFETs.

P2S3: DESAT

[illegible]

P254: DRIVER AND 2 GATES

U704

+5V 15 VCC1 5 VDD_P254
 IN+_P254 10 IN+ 4 OUT+_P254
 IN-_P254 11 IN- 6 OUTL_P254
 EN# 14 RST/EN 7 CLMP_P254
 FLT 13 FLT 2 DX_P254
 RDY 12 RDY DESAT
 P10_P254 16 TEST 3 COM_P254
 GND 9 GND1 1 P1_P254
 GND2 8 VEE_P254
 NSI6651ASC-Q15WR

Q709 PMV30ENEAR

CG_P254 1 VEE1_P254 3 VEE2_P254 4 CLMP_P254 5

Q710 PMV30ENEAR

CG_P254 1 VEE1_P254 3 VEE2_P254 4 CLMP_P254 5

Q711 PMV30ENEAR

CG_P254 1 VEE1_P254 3 VEE2_P254 4 CLMP_P254 5

Components:

- R740 0R
- R741 0R
- R742 1k
- C721 10u 50V
- C722 100n 50V
- C723 22u 25V
- C724 22u 25V
- C725 100n 50V
- R743 3.75R
- R744 2.5R
- R745 10k
- D720 82K84-820.215
- D721 82K84-84V7
- R746 0.5R
- D722 PMEG4010CEJ
- C726 1u 50V
- R747 4.7R
- R748 22R
- R749 3.75R
- R750 2.5R
- R751 10k
- D723 82K84-820.215
- D724 82K84-84V7
- R752 0.5R
- D725 PMEG4010CEJ
- C727 1u 50V
- R753 4.7R
- R754 22R
- R755 150R

Per device: Ron/Roff: 0.5R Kelvin R to the COM star. 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device. loop <= 1 nH.

[illegible]

P2S4: SHORT-CIRCUIT BOOSTER

Figure 10 shows three circuit diagrams for the DESAT pin connections. The top-left diagram shows the connection of the DESAT pin to COM_P254, VDD_P254, and NDB_P254. The top-right diagram shows the connection of the DESAT pin to NDB_P254, VDD_P254, and NDX_P254. The bottom diagram shows the connection of the DESAT pin to D51_P254, D52_P254, and D51_P254.

Top-Left Diagram: The DESAT pin is connected to COM_P254 through a 10k resistor (R768). It is also connected to VDD_P254 through a 100k resistor (R769). The NDB_P254 pin is connected to the DESAT pin through a 100R resistor (R770).

Top-Right Diagram: The DESAT pin is connected to NDB_P254 through a 100R resistor (R770). It is also connected to VDD_P254 through a 100k resistor (R769). The NDX_P254 pin is connected to the DESAT pin through a 100R resistor (R771).

Bottom Diagram: The DESAT pin is connected to D51_P254 through a 100R resistor (R772). It is also connected to D52_P254 through a 100R resistor (R773). The D51_P254 pin is connected to the DESAT pin through a 100R resistor (R774).

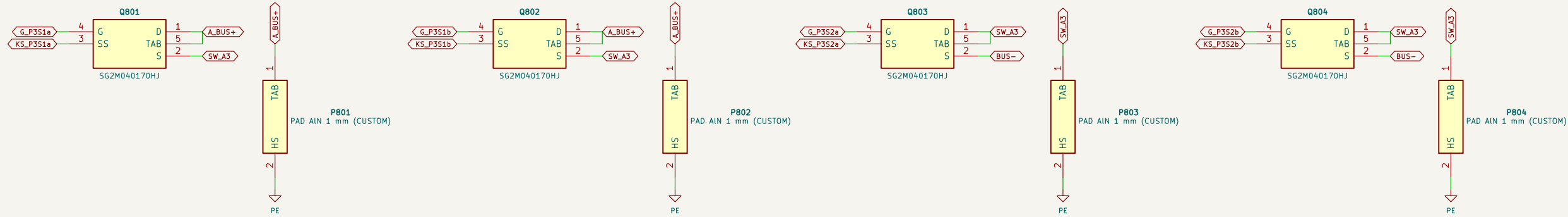
LEG B: LAYOUT RULES

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE baid

File: 07_ph2_gdb.kicad_sch

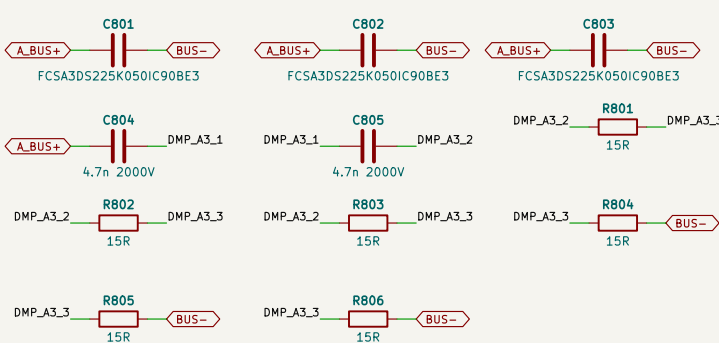
Size: A2	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 8/23

PHASE 3 LEG A: S1 (TOP) / S2 (BOTTOM), 2 X SG2M040170HJ EACH



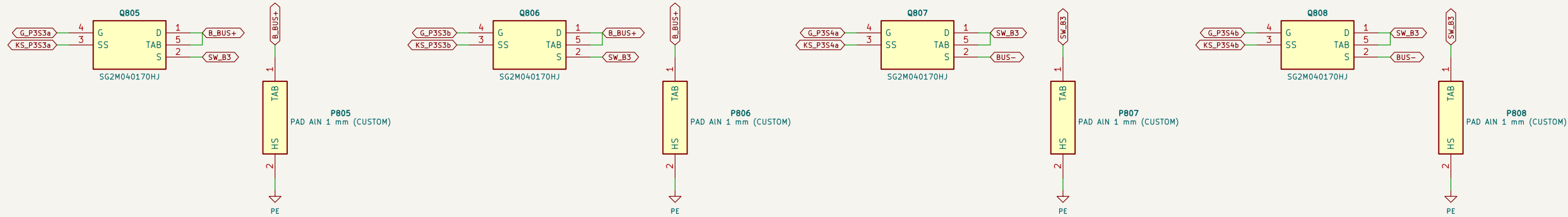
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG A DECOUPLING + RC DAMPER



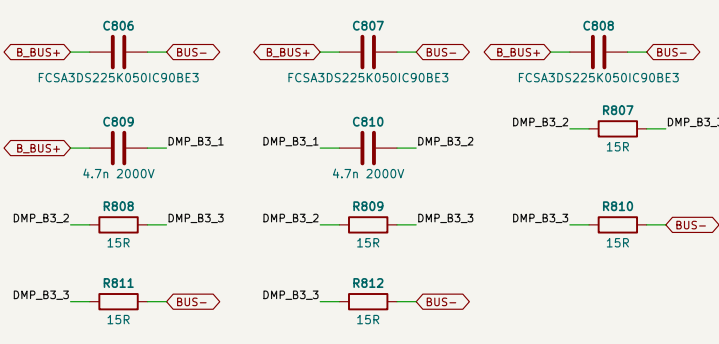
3 x FCSA3DS225 at the S1/S2 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 LEG B: S3 (TOP) / S4 (BOTTOM), 2 X SG2M040170HJ EACH



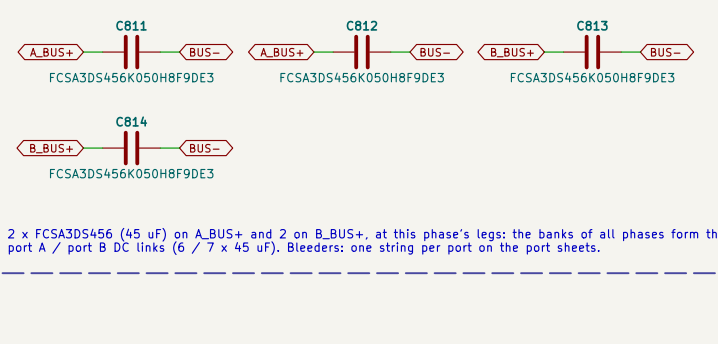
Kelvin source (pin 3) of each device to its own driver COM star (gate-drive sheet). Each tab on its own
AlN ceramic (aluminium nitride) pad to the EARTHED heatsink (basic HV-PE, B2). LAYOUT (IC-22): milled slot between drain pin 1 and
pins 2-4; equal drain / source busbar paths for the two devices of a switch.

PHASE 3 LEG B DECOUPLING + RC DAMPER



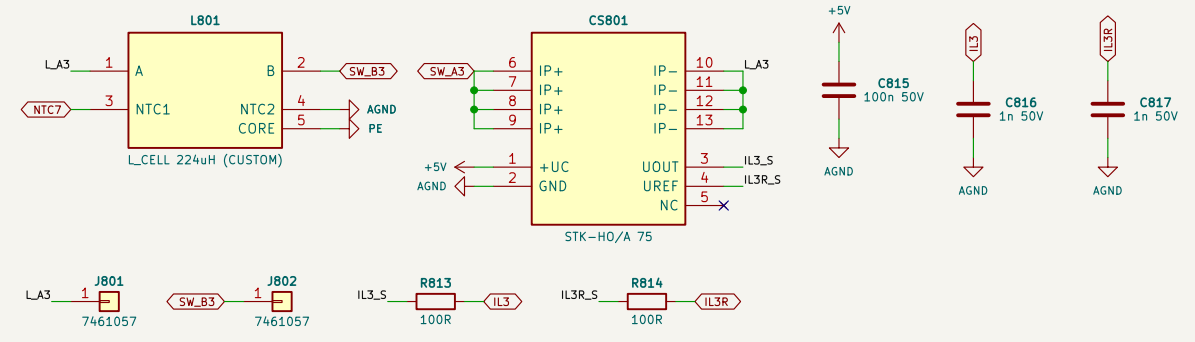
3 x FCSA3DS225 at the S3/S4 pins: commutation loop <= 20 nH, bank <= 15 nH from the leg (LAYOUT, leg deck).
Damper (PVR-02): 2 x (3 x 15R CRCW2512-HP) + 2 x 4.7n 2 kV C0G in series, loop at the pins <= 3 nH.

PHASE 3 SHARE OF THE PORT BANKS



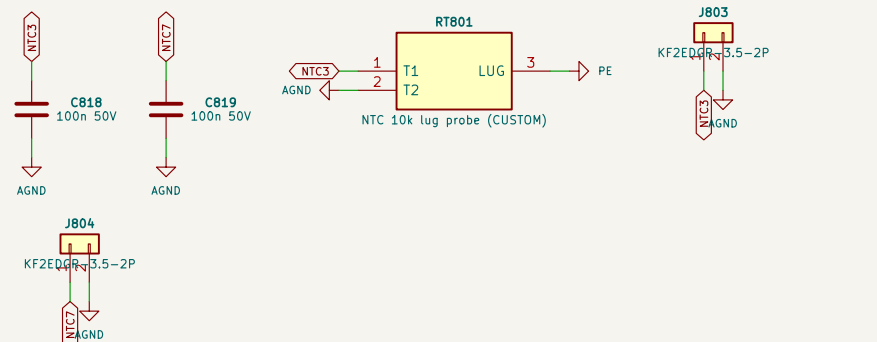
2 x FCSA3DS456 (45 uF) on A.BUS+ and 2 on B.BUS+, at this phase's legs: the banks of all phases form the
port A / port B DC links (6 / 7 x 45 uF). Bleeders: one string per port on the port sheets.

PHASE 3 INDUCTOR (CHASSIS, CUSTOM) AND CURRENT SENSOR



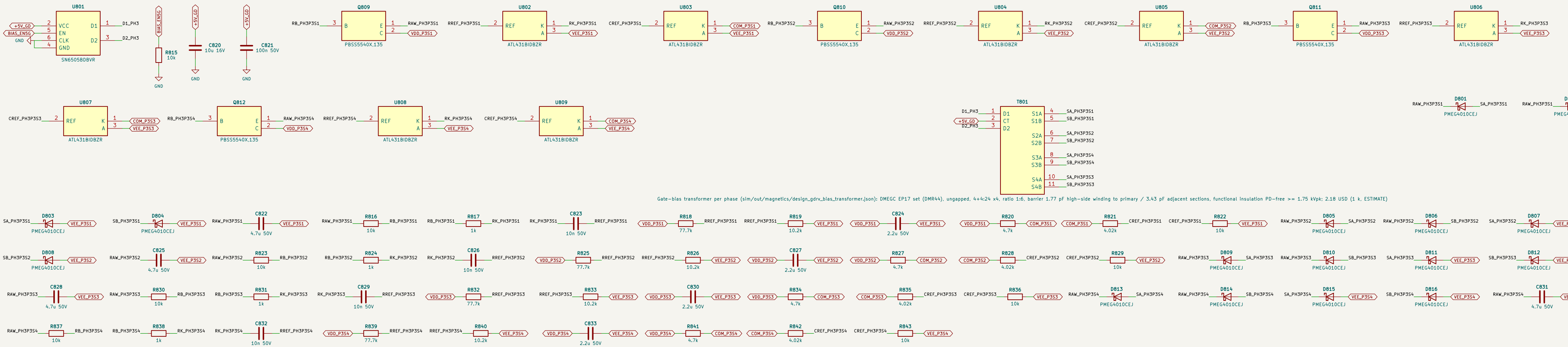
Lead A from SW.A3 passes the STK-HO/A 75 U-primary (pins 6-9 in, 10-13 out; += SW.A -> SW.B) to stud
LA3: lead B on stud SW.B3, IL3 = 2.5 V + 10.667 mV/A (Vref 2.48-2.52 V), 0.5-0.5 V over +/-187.5 A;
100R = 1 nF at the source, IL3R = the sensor's own tref (pin 6), 100R + 1 nF (PVC-CTL references its trip
window to IL. Primary at the switch node: functional barrier (4 kV rms, 8 kV impulse).

PHASE 3 NTC INPUTS (NTC TO AGND, BIASED ON THE CONTROL BOARD)



NTC3 = heatsink section of phase 3: lug probe on the EARTHED heatsink, >= 2.5 kV rms lead-to-lug (R-10).
NTC7 = inductor winding NTC (functional to the winding), 100 nF at the header against switch-node noise.

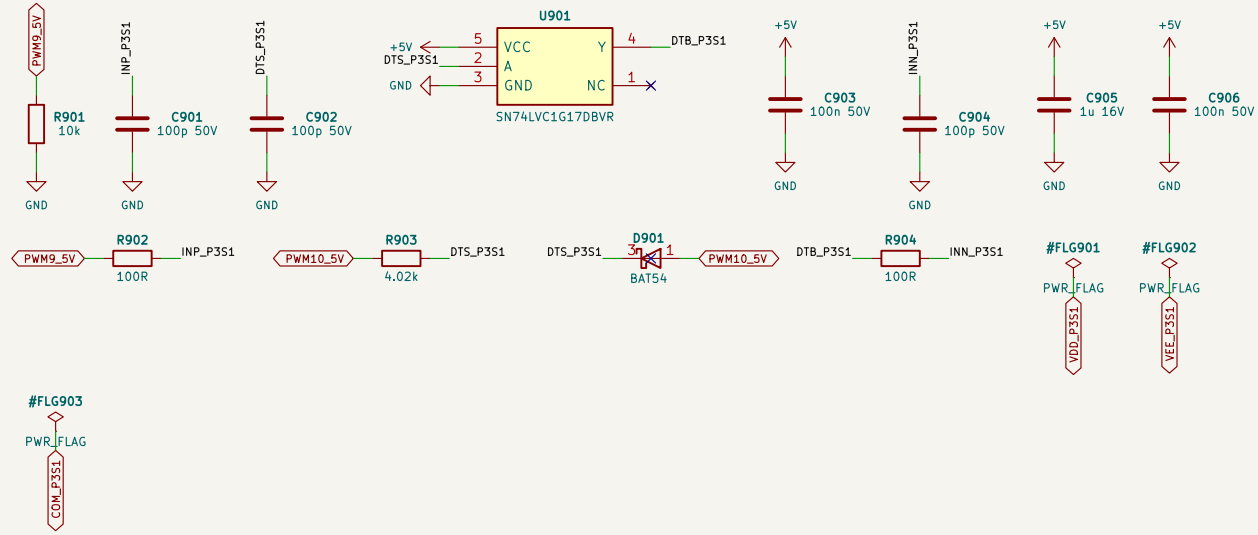
PH3: GATE BIAS (PER PHASE)



Gate-bias transformer per phase (sim/out/magnetics/design_gdrv_bias_transformer.json): DMECC EP17 set (DMR44), ungapped, 4+4:2k v4, ratio 1:6, barrier 1.77 pF high-side winding to primary / 3.43 pF adjacent sections, functional insulation PD-free >= 1.75 kVpk; 2.18 USD (1 k, ESTIMATE)

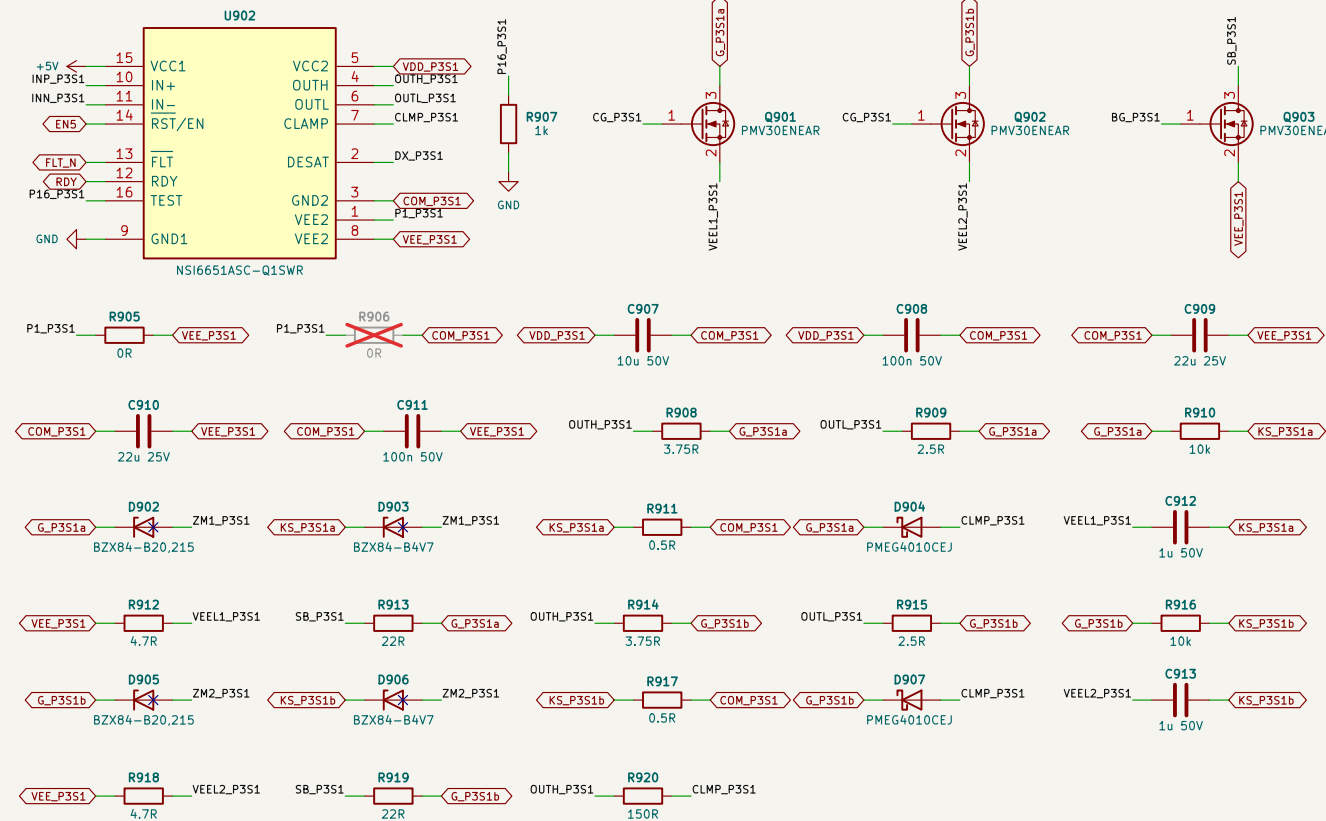
SN6505B push-pull 420 kHz from +5V_GD, BIAS_EN high = on; T.BIAS4 custom, functional insulation
(PD-free >= 1.75 kVpk, <= 5 pF per secondary). Per channel: bridge, PNP regulator VDD-VEE, ATL431 COM split.

P3S1: LOGIC SIDE



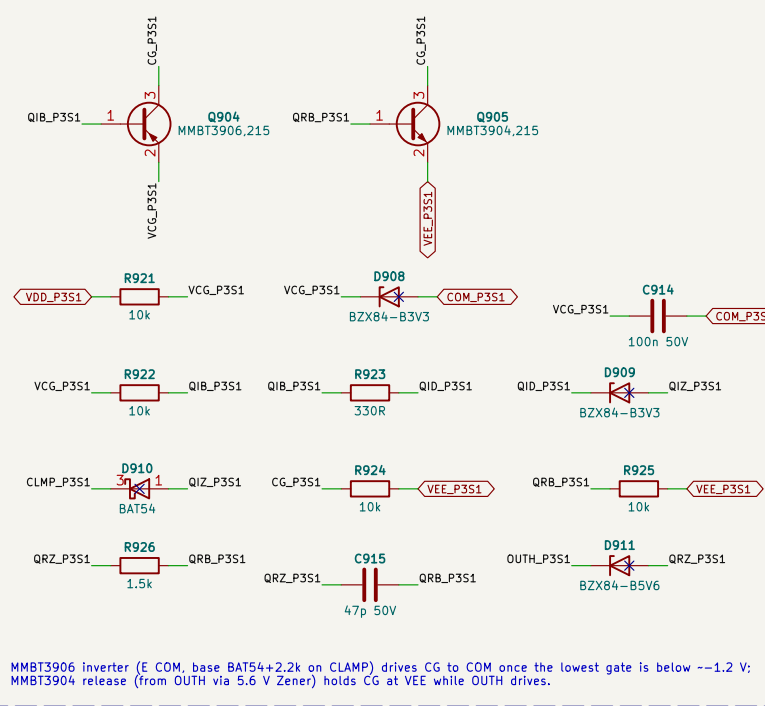
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(Interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S1: DRIVER AND 2 GATES



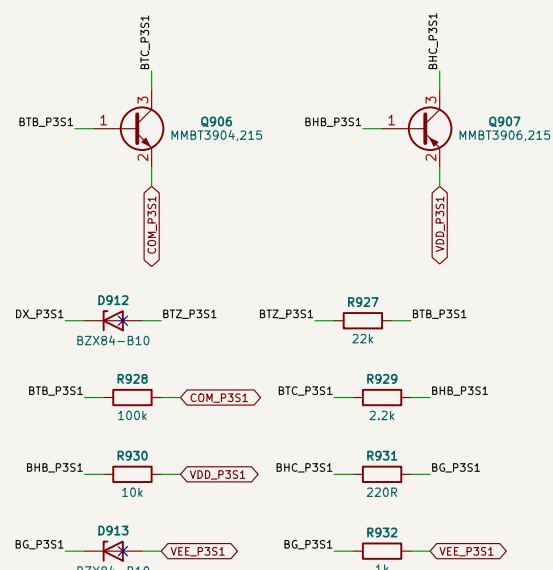
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S1: MILLER CLAMP CONTROL



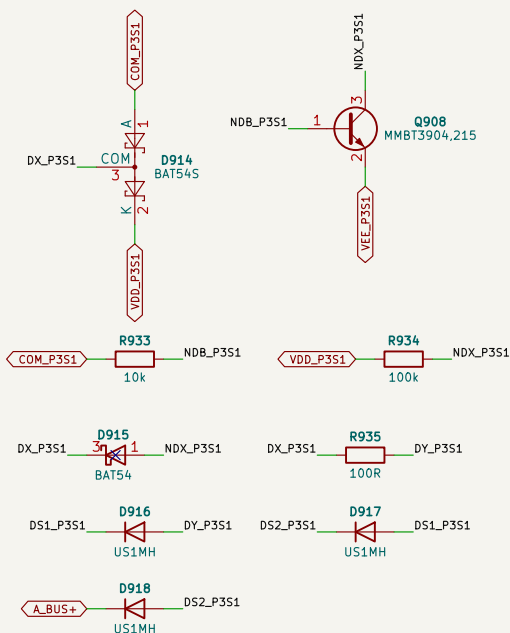
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S1: SHORT-CIRCUIT BOOSTER



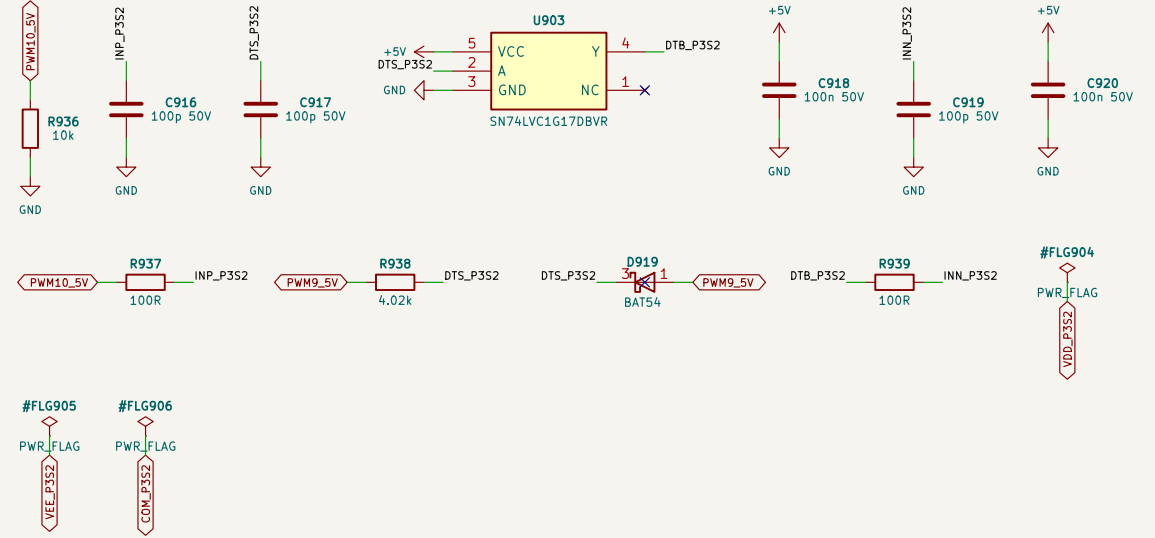
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S1: DESAT



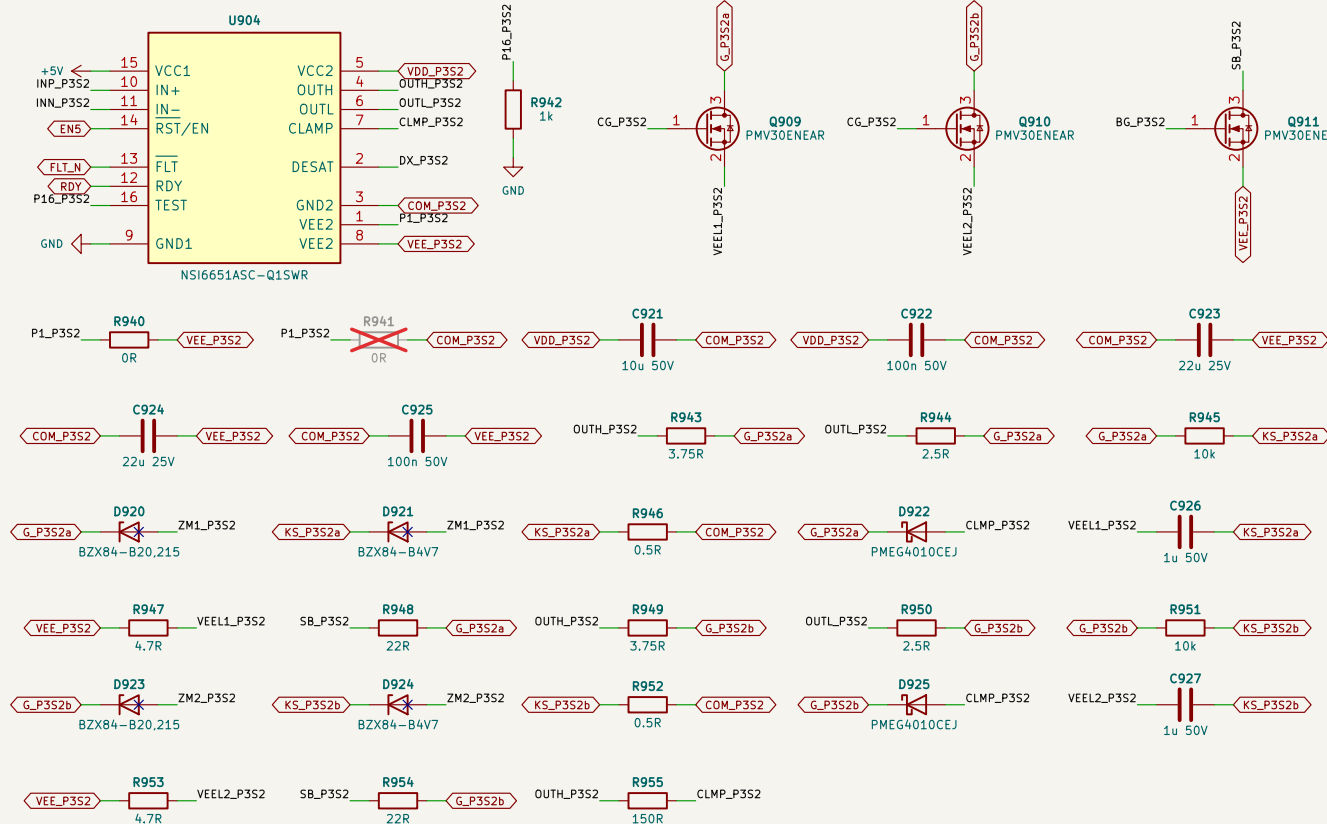
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

P3S2: LOGIC SIDE



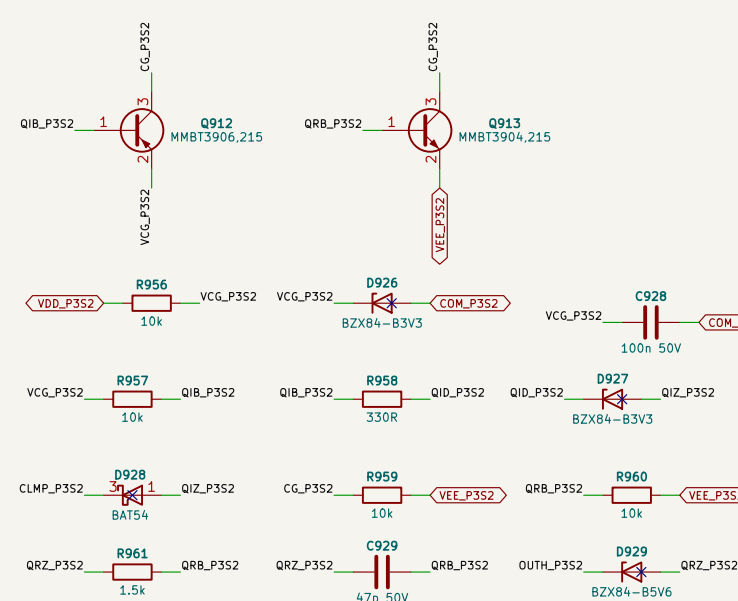
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p
(interlock + min dead time = max(firmware, hardware)). Bias PG (active low) pulls RDY low.

P3S2: DRIVER AND 2 GATES



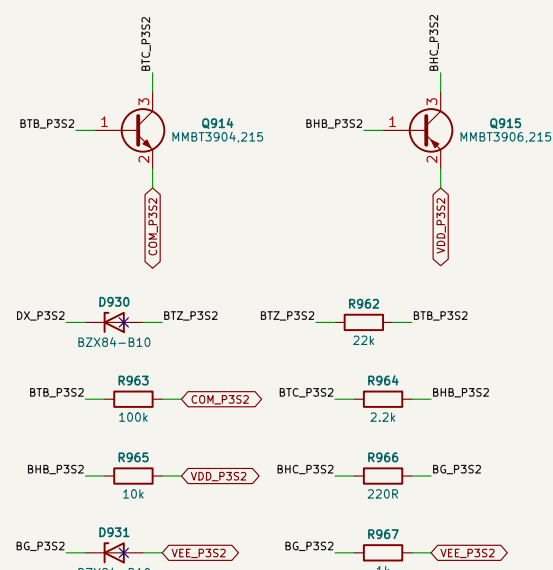
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net.
Miller hold: N-MOSFET clamp (PMV30ENEAE) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S2: MILLER CLAMP CONTROL



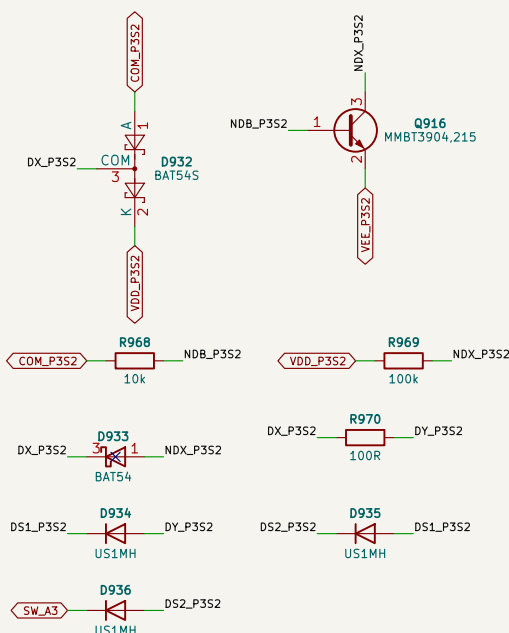
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below --1.2 V;
MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S2: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V;
per device AQ3400A + 22R pull the gate to VEE; BG holds ~3 us (1k + FET gates).

P3S2: DESAT

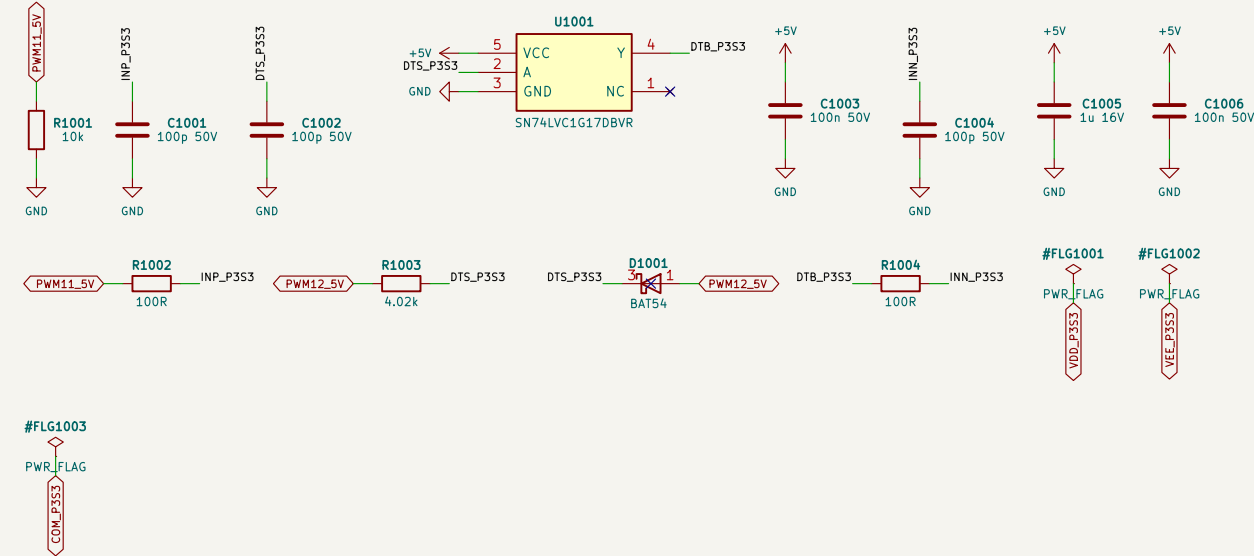


DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain.
BAT54S keeps DX between COM and VDD against dv/dt through the string; Cdk footprint DNP.

LEG A: LAYOUT RULES

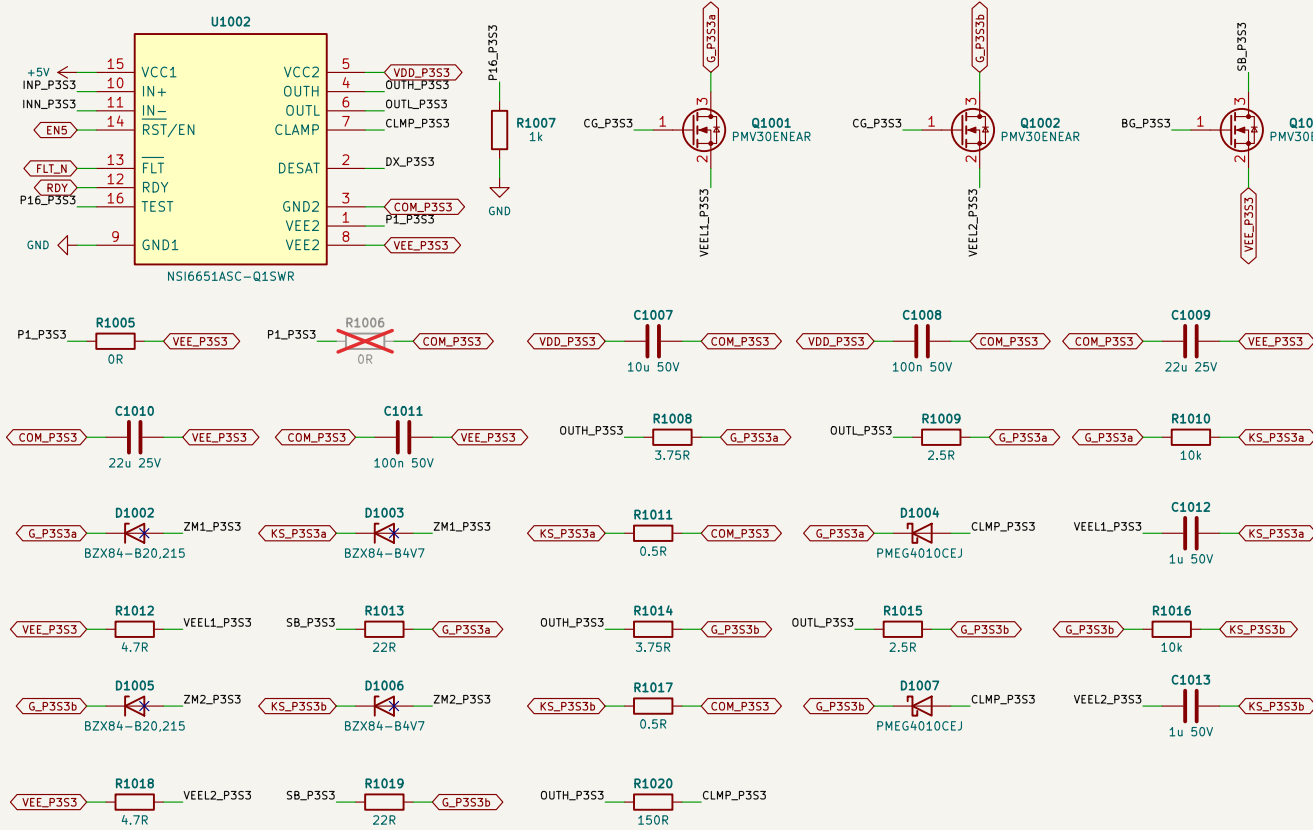
Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + C.VL at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PD1 (IC-22). Miller hold (opposite, zero): die +1.41 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

P3S3: LOGIC SIDE



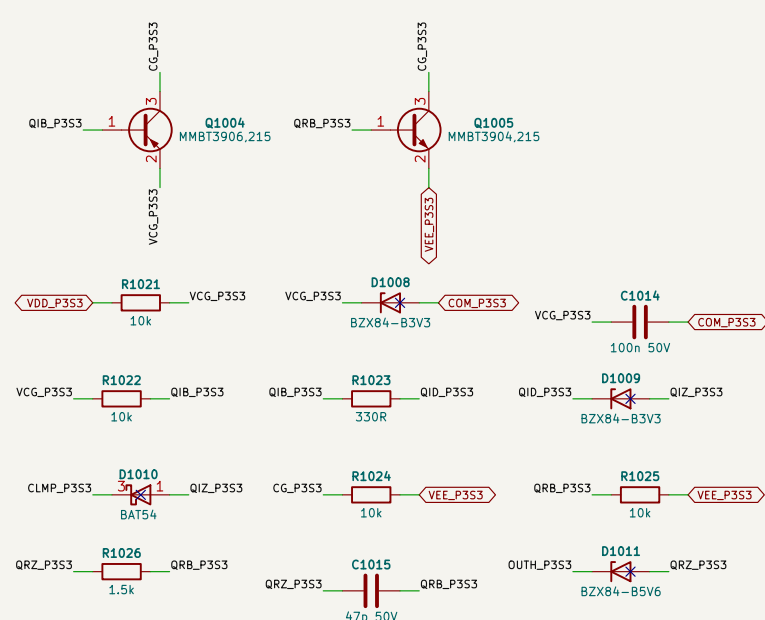
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S3: DRIVER AND 2 GATES



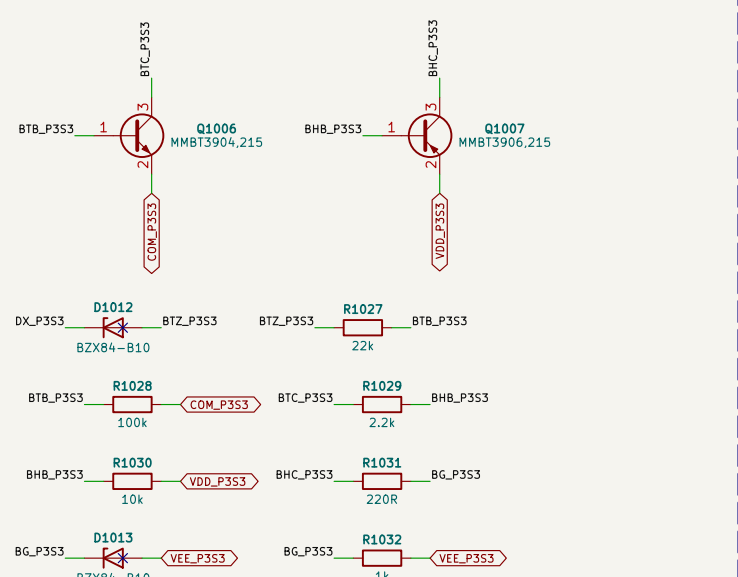
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S3: MILLER CLAMP CONTROL



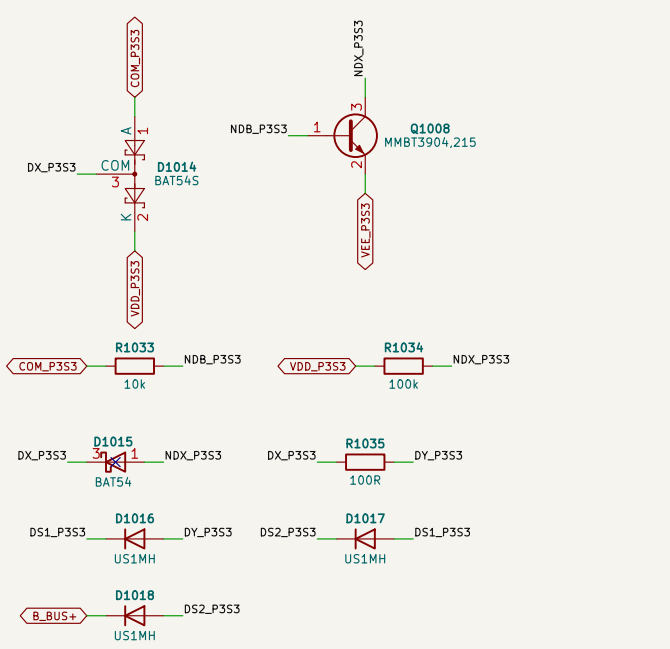
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S3: SHORT-CIRCUIT BOOSTER



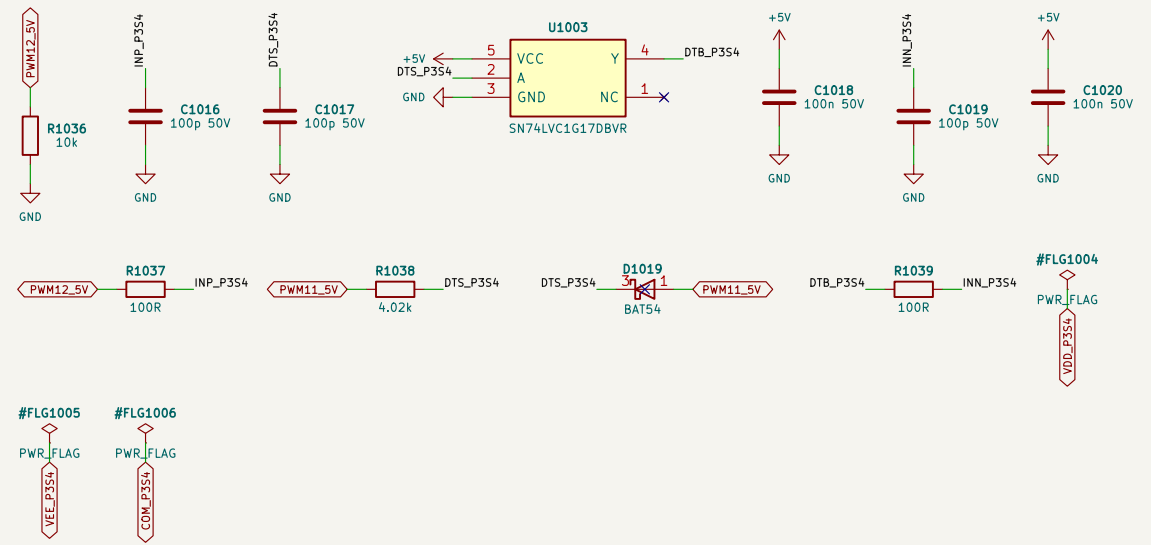
DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S3: DESAT



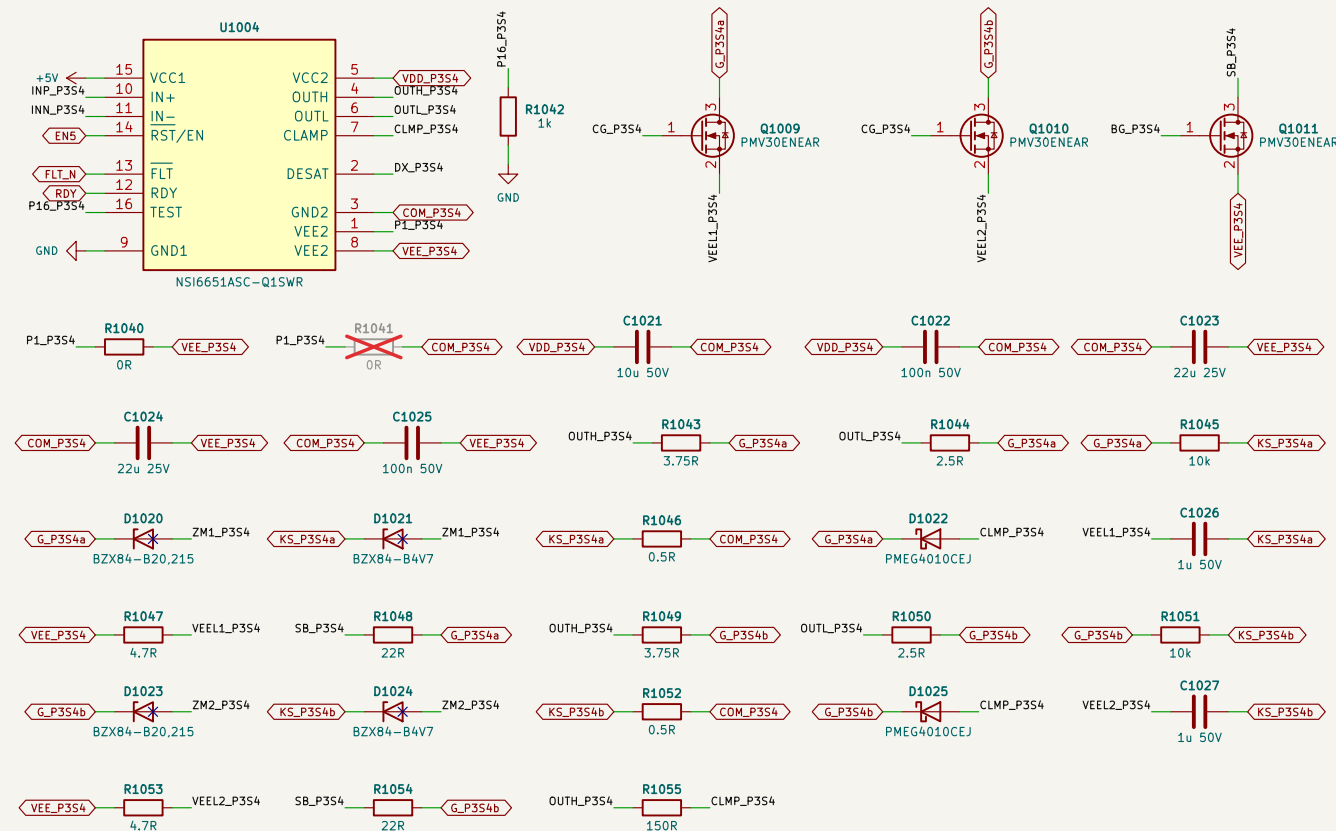
DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Clik footprint DNP.

P3S4: LOGIC SIDE



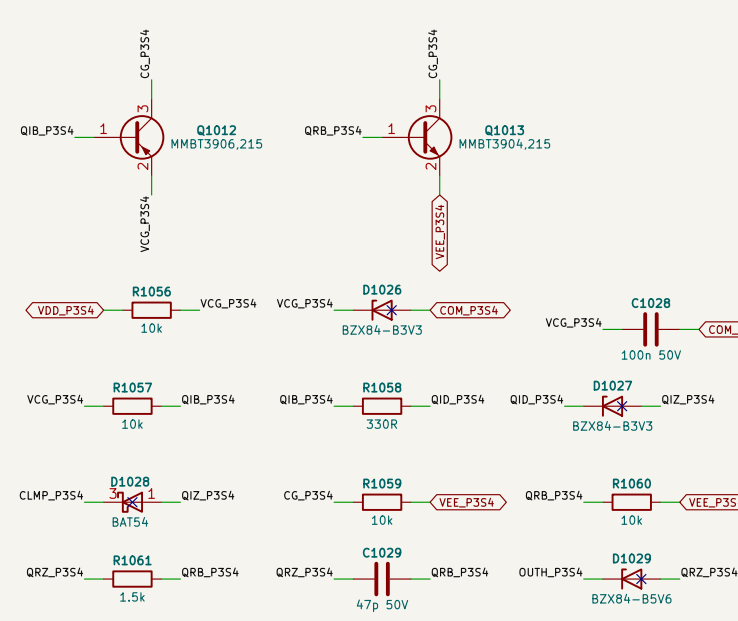
IN+ = own PWM; IN- = complementary switch, falling edge stretched 4.02k/100p (interlock + min dead time = max(firmware, hardware)); Bias PG (active low) pulls RDY low.

P3S4: DRIVER AND 2 GATES



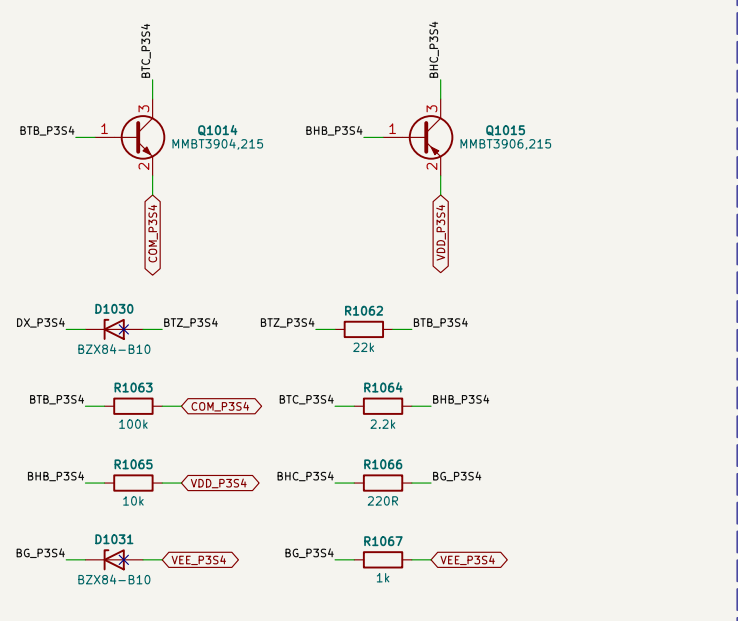
Per device: Ron/Roff, 0.5R Kelvin R to the COM star, 10k and Zeners (about +21 V / -5.7 V) on its own Kelvin net. Miller hold: N-MOSFET clamp (PMV30ENEA) gate-pin to a local VEE (1u on the Kelvin pin). LAYOUT: FET + 1u at the device, loop <= 1 nH.

P3S4: MILLER CLAMP CONTROL



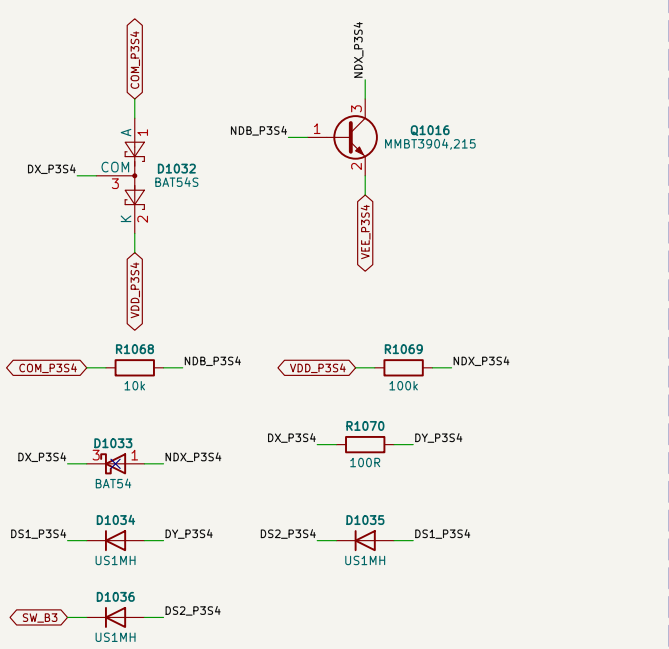
MMBT3906 inverter (E COM, base BAT54+2.2k on CLAMP) drives CG to COM once the lowest gate is below -1.2 V; MMBT3904 release (from OUTH via 5.6 V Zener) holds CG at VEE while OUTH drives.

P3S4: SHORT-CIRCUIT BOOSTER



DESAT > 10.3-10.9 V (above the 9.8 V maximum trip) -> MMBT3904 -> MMBT3906 charges BG to VEE + 10 V per device A03400A + 22R pull the gate to VEE; BG holds -3 us (1k + FET gates).

P3S4: DESAT



DESAT pin (internal 500 uA blanking source, trip 8.5-9.8 V) - Rs = 3 x US1M - drain. BAT54S keeps DX between COM and VDD against dv/dt through the string; Clik footprint DNP.

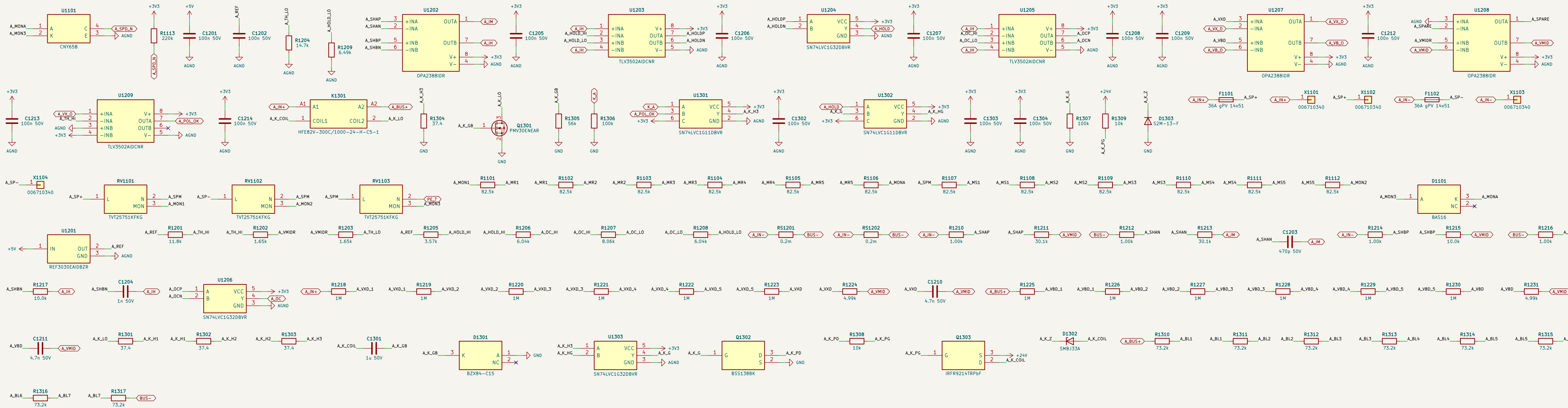
LEG B: LAYOUT RULES

Gate and Kelvin of each device as coupled pairs of equal length (+/-10 %) from the COM star; Ron, Roff, Kelvin R, 10k, Zeners at each device. Clamp FET + CxV at its device's gate-Kelvin pins, loop <= 1 nH. DESAT string >= 2.5 mm pad gap per diode; board coated to PDI (IC-22); Miller hold (optical, zero): die +1.4 V vs VGS(n) min 1.94 V at 170 C - BENCH TEST. Driver input side and bias primary decoupled to GND locally (CM current).

PORT A (PV SOURCE), D-044 LEAN PORT

Terminal side A_IN+ / A_IN-, bank A_BUS+. No fuse, no precharge; array Isc <= 1.25 x rating.
Contactor opens <= 0.97 kA, held above (port_spec lean). Installation: upstream clears 967-1250 A within 0.26 s.

ON-BOARD SURGE PROTECTION: VARISTOR Y (IC-01, IC-12)



TV125751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/-M, MOV2 +/-M, MOV3 M/PE,I. Up.eff 3793 V pole-PE and pole-pole at In 3 kA (calc. <= 4.0 kV -> reinforced 6 kV / basic 4 kV, D-032); Imax 25 kA. Branch fuses ETI CH14x21 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three links -> CNY65B -> RB1 (sum on the ID line); RB1 = 1 at ms; above 200 V = varistor network open -> CTRL inhibits/stops.

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 13
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /12 Port A sensing and interlocks/
File: 12_leanA_ctrl.kicad_sch

Title: **PV-PWR power board – Port A sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 13/23

	1	2	3	4	5	6	7	8
A								
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E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 14
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /13 Port A coil drives/
File: 13_leanA_drive.kicad_sch

Title: **PV-PWR power board – Port A coil drives**

Size: A3

Date: 2026-10-05

Rev: **A2**

KiCad E.D.A. 10.0.6

Id: 14/23

PORT B (BATTERY SOURCE), D-044 LEAN PORT

Terminal side B_{IN+} / B_{IN-}, bank B_{BUS+}. HIF501 250 A aR in both poles (breaks 1.25–50 kA only).
Contactor opens <= 0.97 kA, held above (port-spec lean). Installation: upstream clears 967–1250 A within 0.26 s.

[illegible]

TV725751 x 3 (615 VDC, thermal link + monitor lead); MOV1 +/M, MOV2 -/M, MOV3 M/PE_T, U_{g,eff} 3793 V pole-PE and pole-pole at in 3 kA (calc., $u = 4.0 \text{ kV} \Rightarrow$ reinforced 6 kV / basic 4 kV, D=032); I_{max} 25 kA. Branch fuses ETI CH14x51 gPV 36 A 30 kA (L/R 2 ms). Monitor loop through both fuses and all three Units $\Rightarrow$ CNY658 $\Rightarrow$ RB1 (sum on the ID line); RB1 = 1 at rest above 200 V = varistor network open $\Rightarrow$ CTRL inhibits/stops.

FUSES (BATTERY-TYPE PORT)

The diagram illustrates the electrical connections for the fuses and associated components in the BATTERY-TYPE PORT section. Key components include:

- Fuses:** R1500, R1501, R1502, R1503, R1504, R1505, R1506, R1507, R1508, R1509, R1510, R1511, R1512, R1513, R1514, R1515, R1516, R1517, R1518, R1519, R1520, R1521, R1522, R1523, R1524, R1525, R1526, R1527, R1528, R1529, R1530, R1531, R1532, R1533, R1534, R1535, R1536, R1537, R1538, R1539, R1540, R1541, R1542, R1543, R1544, R1545, R1546, R1547, R1548, R1549, R1550.
- Capacitors:** C1500, C1501, C1502, C1503, C1504, C1505, C1506, C1507, C1508, C1509, C1510, C1511, C1512, C1513, C1514, C1515, C1516, C1517, C1518, C1519, C1520, C1521, C1522, C1523, C1524, C1525, C1526, C1527, C1528, C1529, C1530, C1531, C1532, C1533, C1534, C1535, C1536, C1537, C1538, C1539, C1540, C1541, C1542, C1543, C1544, C1545, C1546, C1547, C1548, C1549, C1550.
- Integrated Circuits:** U1500, U1501, U1502, U1503, U1504, U1505, U1506, U1507, U1508, U1509, U1510, U1511, U1512, U1513, U1514, U1515, U1516, U1517, U1518, U1519, U1520, U1521, U1522, U1523, U1524, U1525, U1526, U1527, U1528, U1529, U1530, U1531, U1532, U1533, U1534, U1535, U1536, U1537, U1538, U1539, U1540, U1541, U1542, U1543, U1544, U1545, U1546, U1547, U1548, U1549, U1550.
- Other Components:** Q1600, Q1601, Q1602, Q1603, Q1604, Q1605, Q1606, Q1607, Q1608, Q1609, Q1610, Q1611, Q1612, Q1613, Q1614, Q1615, Q1616, Q1617, Q1618, Q1619, Q1620, Q1621, Q1622, Q1623, Q1624, Q1625, Q1626, Q1627, Q1628, Q1629, Q1630, Q1631, Q1632, Q1633, Q1634, Q1635, Q1636, Q1637, Q1638, Q1639, Q1640, Q1641, Q1642, Q1643, Q1644, Q1645, Q1646, Q1647, Q1648, Q1649, Q1650, Q1651, Q1652, Q1653, Q1654, Q1655, Q1656, Q1657, Q1658, Q1659, Q1660, Q1661, Q1662, Q1663, Q1664, Q1665, Q1666, Q1667, Q1668, Q1669, Q1670, Q1671, Q1672, Q1673, Q1674, Q1675, Q1676, Q1677, Q1678, Q1679, Q1680, Q1681, Q1682, Q1683, Q1684, Q1685, Q1686, Q1687, Q1688, Q1689, Q1690, Q1691, Q1692, Q1693, Q1694, Q1695, Q1696, Q1697, Q1698, Q1699, Q1700, Q1701, Q1702, Q1703, Q1704, Q1705, Q1706, Q1707, Q1708, Q1709, Q1710, Q1711, Q1712, Q1713, Q1714, Q1715, Q1716, Q1717, Q1718, Q1719, Q1720, Q1721, Q1722, Q1723, Q1724, Q1725, Q1726, Q1727, Q1728, Q1729, Q1730, Q1731, Q1732, Q1733, Q1734, Q1735, Q1736, Q1737, Q1738, Q1739, Q1740, Q1741, Q1742, Q1743, Q1744, Q1745, Q1746, Q1747, Q1748, Q1749, Q1750, Q1751, Q1752, Q1753, Q1754, Q1755, Q1756, Q1757, Q1758, Q1759, Q1760, Q1761, Q1762, Q1763, Q1764, Q1765, Q1766, Q1767, Q1768, Q1769, Q1770, Q1771, Q1772, Q1773, Q1774, Q1775, Q1776, Q1777, Q1778, Q1779, Q1780, Q1781, Q1782, Q1783, Q1784, Q1785, Q1786, Q1787, Q1788, Q1789, Q1790, Q1791, Q1792, Q1793, Q1794, Q1795, Q1796, Q1797, Q1798, Q1799, Q1800, Q1801, Q1802, Q1803, Q1804, Q1805, Q1806, Q1807, Q1808, Q1809, Q1810, Q1811, Q1812, Q1813, Q1814, Q1815, Q1816, Q1817, Q1818, Q1819, Q1820, Q1821, Q1822, Q1823, Q1824, Q1825, Q1826, Q1827, Q1828, Q1829, Q1830, Q1831, Q1832, Q1833, Q1834, Q1835, Q1836, Q1837, Q1838, Q1839, Q1840, Q1841, Q1842, Q1843, Q1844, Q1845, Q1846, Q1847, Q1848, Q1849, Q1850, Q1851, Q1852, Q1853, Q1854, Q1855, Q1856, Q1857, Q1858, Q1859, Q1860, Q1861, Q1862, Q1863, Q1864, Q1865, Q1866, Q1867, Q1868, Q1869, Q1870, Q1871, Q1872, Q1873, Q1874, Q1875, Q1876, Q1877, Q1878, Q1879, Q1880, Q1881, Q1882, Q1883, Q1884, Q1885, Q1886, Q1887, Q1888, Q1889, Q1890, Q1891, Q1892, Q1893, Q1894, Q1895, Q1896, Q1897, Q1898, Q1899, Q1900, Q1901, Q1902, Q1903, Q1904, Q1905, Q1906, Q1907, Q1908, Q1909, Q1910, Q1911, Q1912, Q1913, Q1914, Q1915, Q1916, Q1917, Q1918, Q1919, Q1920, Q1921, Q1922, Q1923, Q1924, Q1925, Q1926, Q1927, Q1928, Q1929, Q1930, Q1931, Q1932, Q1933, Q1934, Q1935, Q1936, Q1937, Q1938, Q1939, Q1940, Q1941, Q1942, Q1943, Q1944, Q1945, Q1946, Q1947, Q1948, Q1949, Q1950, Q1951, Q1952, Q1953, Q1954, Q1955, Q1956, Q1957, Q1958, Q1959, Q1960, Q1961, Q1962, Q1963, Q1964, Q1965, Q1966, Q1967, Q1968, Q1969, Q1970, Q1971, Q1972, Q1973, Q1974, Q1975, Q1976, Q1977, Q1978, Q1979, Q1980, Q1981, Q1982, Q1983, Q1984, Q1985, Q1986, Q1987, Q1988, Q1989, Q1990, Q1991, Q1992, Q1993, Q1994, Q1995, Q1996, Q1997, Q1998, Q1999, Q2000, Q2001, Q2002, Q2003, Q2004, Q2005, Q2006, Q2007, Q2008, Q2009, Q2010, Q2011, Q2012, Q2013, Q2014, Q2015, Q2016, Q2017, Q2018, Q2019, Q2020, Q2021, Q2022, Q2023, Q2024, Q2025, Q2026, Q2027, Q2028, Q2029, Q2030, Q2031, Q2032, Q2033, Q2034, Q2035, Q2036, Q2037, Q2038, Q2039, Q2040, Q2041, Q2042, Q2043, Q2044, Q2045, Q2046, Q2047, Q2048, Q2049, Q2050, Q2051, Q2052, Q2053, Q2054, Q2055, Q2056, Q2057, Q2058, Q2059, Q2060, Q2061, Q2062, Q2063, Q2064, Q2065, Q2066, Q2067, Q2068, Q2069, Q2070, Q2071, Q2072, Q2073, Q2074, Q2075, Q2076, Q2077, Q2078, Q2079, Q2080, Q2081, Q2082, Q2083, Q2084, Q2085, Q2086, Q2087, Q2088, Q2089, Q2090, Q2091, Q2092, Q2093, Q2094, Q2095, Q2096, Q209

HPE501/000B100-250 in both poles: breaks 1.25-50 kA only (partial range)
the part OC trip and the contactor keep it out of 250 A-1.25 kA.

	1	2	3	4	5	6	7	8
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B								
C								
D								
E								
F								

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 16
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /14 Port B sensing and interlocks/
File: 14_leanB_ctrl.kicad_sch

Title: **PV-PWR power board – Port B sensing and interlocks**

Size: A3

Date: 2026-10-05

Rev: **A2**

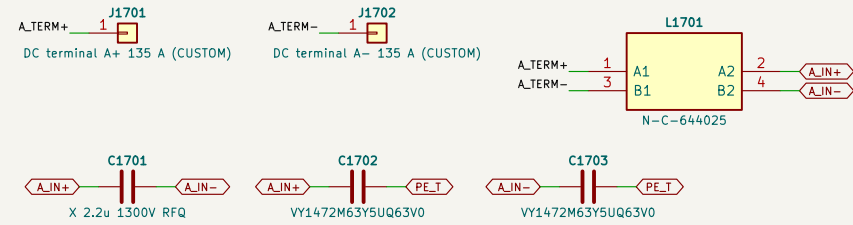
KiCad E.D.A. 10.0.6

Id: 16/23

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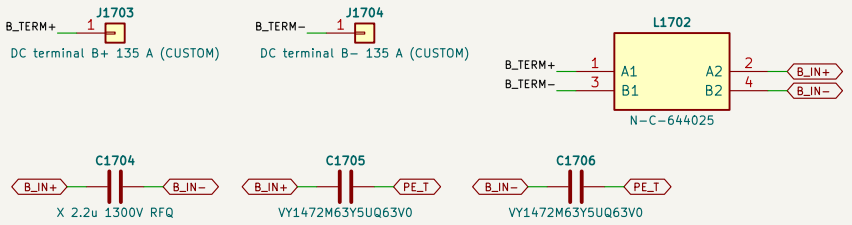
Not bench-validated. Values read from sim/out/*.json; checks: design_check() Sheet 17 Schematics authored by Chinmoy Bhuyan Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic		
Sheet: /15 Port B coil drives/ File: 15_leanB_drive.kicad_sch		
Title: PV-PWR power board – Port B coil drives		
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6	Id: 17/23	

PORT A TERMINALS AND EMI FILTER



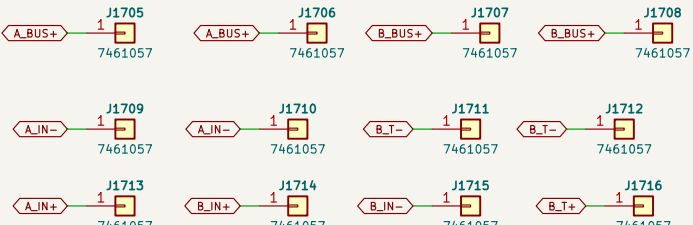
PV: chassis terminals A_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps A_IN+ by short links.

PORT B TERMINALS AND EMI FILTER



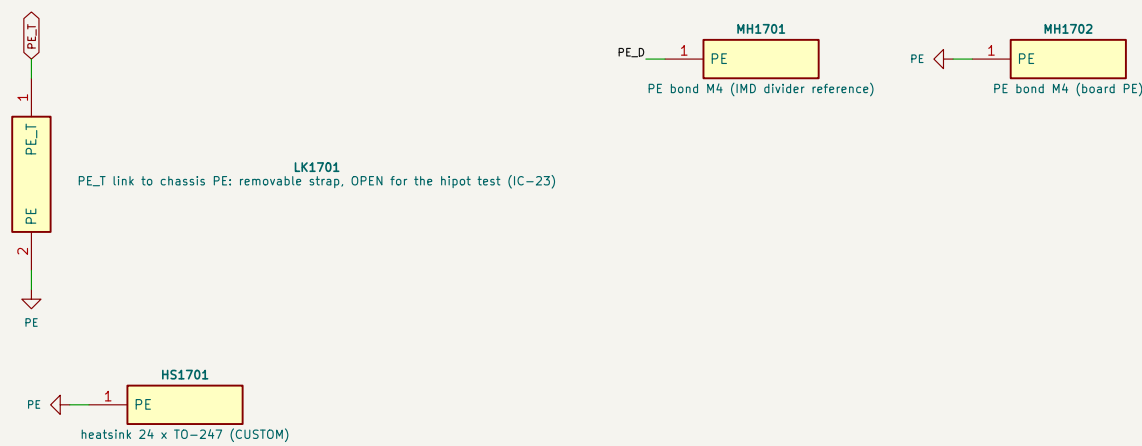
battery: chassis terminals B_TERM+/- (135 A port), both conductors once through the N-C-644025 ring (CM, 150 kHz-30 MHz, ARCHITECTURE 8.1), X 2.2 uF 1300 V at the varistor network, Y1 4.7 nF pole-PE_T (removable PE link for the hipot). Board taps B_IN+ / B_IN- / B_T+ by short links.

BOARD ENTRIES (WURTH REDCUBE PRESS-FIT M3, 100 A EACH)



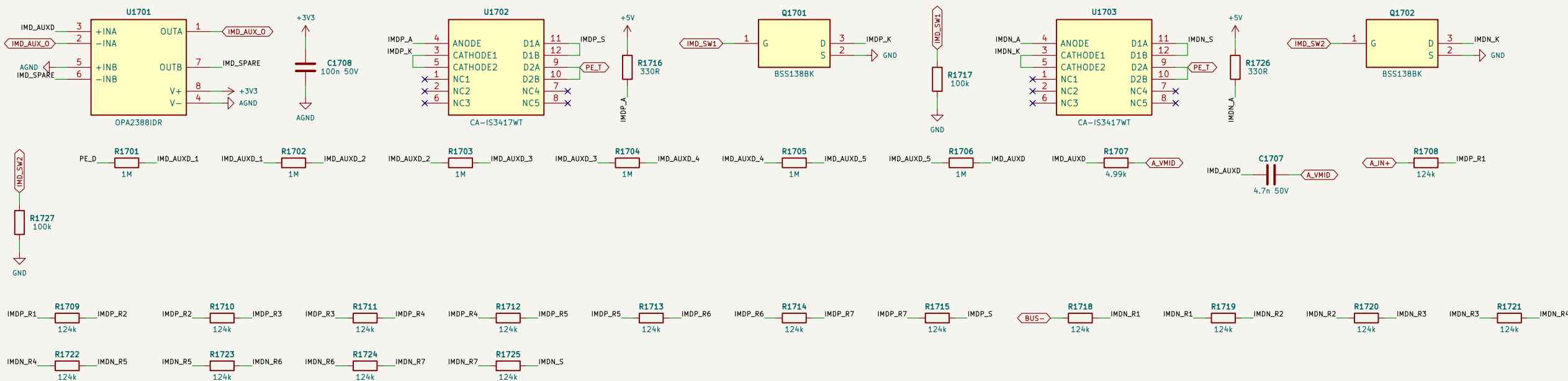
Bus entries from the contactors / fuses: A_BUS+, B_BUS+, A_IN- and B_T- (into the shunts), two studs each; taps A_IN+, B_IN+, B_IN-, B_T+ (varistors, dividers, IMD, precharge, aux input), one each.

PE: LINKS, HEATSINK



PE_T (Y capacitors, MOV3, IMD strings) reaches PE through a removable strap (open for the hipot, IC-23); PE_D = IMD divider bond; the heatsink is earthed at two points.

INSULATION MONITOR (PORT.LEAN_IMD)



IMD_SW1 = string A+ -> PE, IMD_SW2 = string PE -> BUS- (high = connected); VPE = PE divider against BUS- (VMID of port A).

Not bench-validated. Values read from sim/output/*.json; checks: design_check()

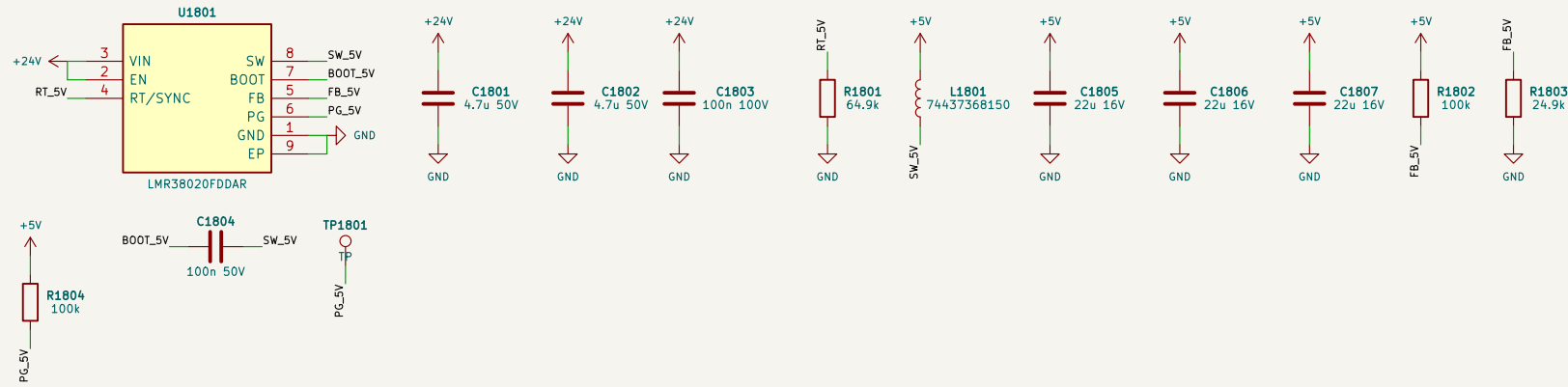
Sheet 18
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /15 Terminals, EMI filter, insulation test/
File: 15_terminals_imd.kicad_sch

Title: PV-PWR power board - Terminals, EMI filter, insulation test

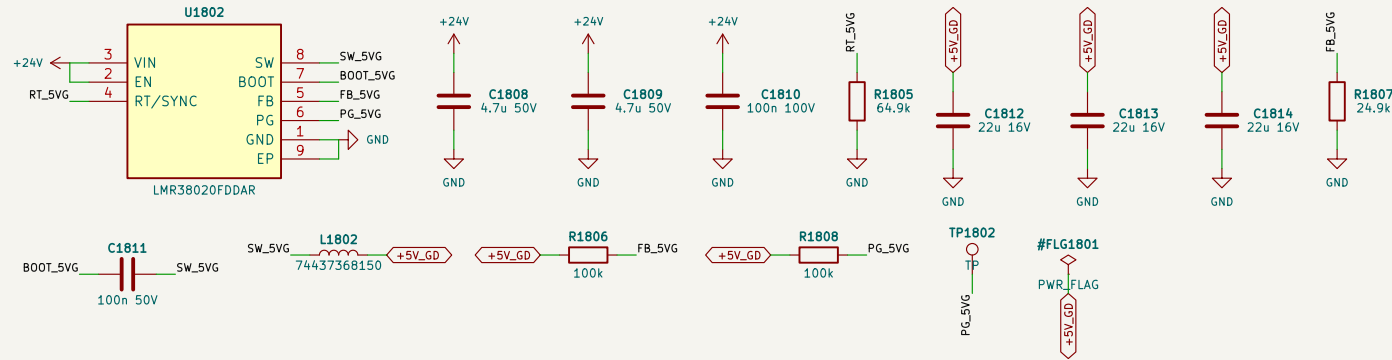
Size: A3 Date: 2026-10-05 Rev: A2
KiCad E.D.A. 10.0.6 Id: 18/23

24 V -> +5V (LMR38020, 400 KHZ, 2 A)



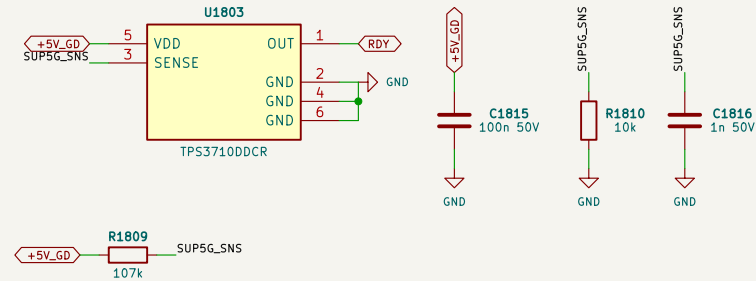
NSI6651 inputs, AHCT, sensors, references, PC +5V, gate bias of phases 1-2, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

24 V -> +5V_GD (LMR38020, 400 KHZ, 2 A)



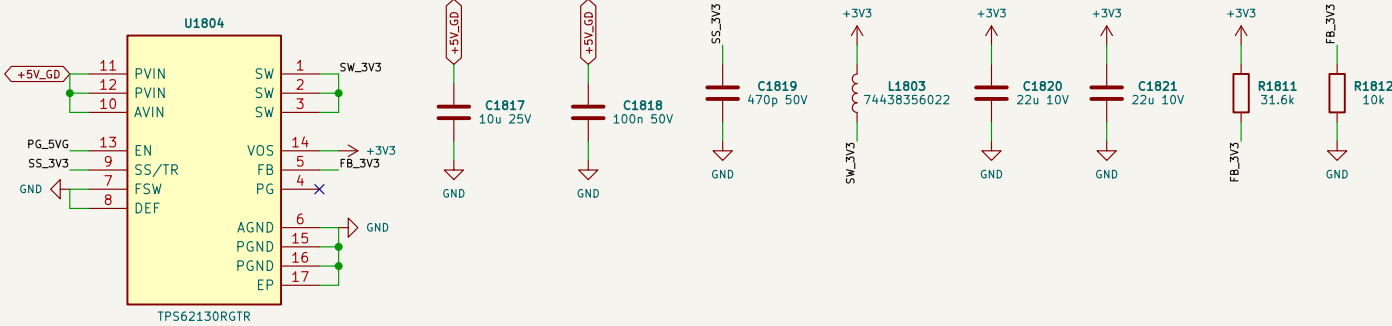
gate bias of phases 3 only, ctrlC2000 circuit (SNVSC40E table 9-1, 400 kHz / 5 V / 2 A): 15 uH, 3 x 22 uF, RT 64.9k, FB 100k / 24.9k 0.1 % -> 4.93-5.10 V. Load in design_check (<= 80 % of 2 A).

+5V_GD SUPERVISOR ON RDY



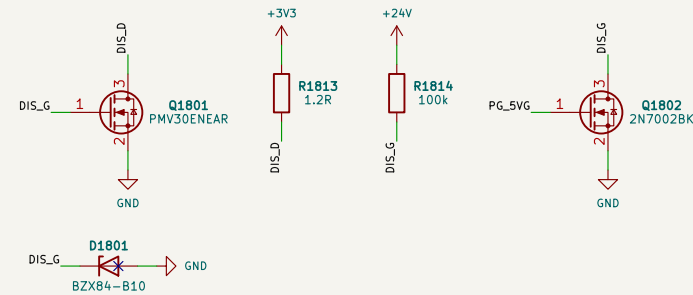
TPS3710 107k/10k as on +5V: RDY low while the bias rail of phases 3 is below 4.52-4.69 V.

+5V_GD -> +3V3 (TPS62130, 2.5 MHZ) WITH A CONTROLLED RAMP



EN = PG_5VG, FB 31.6k / 10k 0.1 % -> 3.240-3.433 V (>= 3.20 V for the control board's reference), SS/TR 470p COG; ramp-up 12-16 mV/us (F28003x VDDIO 8-100 mV/us). Ports' logic, RDY / MOV_OK pull-ups, PC +3V3 (control <= 157 mA).

+3V3 DISCHARGE (RAMP-DOWN WHEN THE RAILS FAIL)



PG_5VG low (+5V_GD < 90-94 %, i.e. the live 24 V is gone) disables the TPS62130 and lets the 100k from +24V (10 V Zener) turn on PMV30ENEA + 1.2R: +3V3 falls at >= 20 mV/us through the BOR region (F28003x SRVDDIO-DN).

SELV 24 V TO THE CONTROL BOARD (J2)



AUX-T1 reinforced winding SELV_24V / SELV_0V, cross-regulated 20.3-29.8 V (aux75_spec); full fan speed needs 24.5 V. Full fan speed is only guaranteed while the converter is switching. Fan buck (12-36 V in) + SELV logic on the control board; >= 10 mm from live copper; nothing else of the SELV domain is on this board.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()

Sheet 19

Schematics authored by Chinmoy Bhuyan

Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /16 Live 5 V / 3.3 V, SELV connector/

File: 16_supplies.kicad_sch

Title: PV-PWR power board - Live 5 V / 3.3 V, SELV connector

Size: A3

Date: 2026-10-05

Rev: A2

KiCad E.D.A. 10.0.6

Id: 19/23

TAP A_IN+



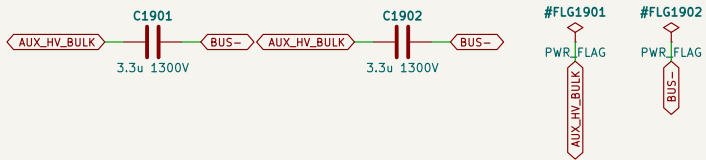
2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

TAP B_T+



2 per tap AHEAD of the fuse (AC10 pulse-rated, 20 R = the smallest E24 value inside AX-01 / 02: hot-plug I2t 18.1 % of the fuse melting I2t (rule 20 %), film surge 1218 V); 4.14 W each at 200 V / 75 W Fuse: (30 W case, unchanged) per-port fuse 1000 VDC gPV behind 44R: fault current ≤ 25 A (breaking 20 kA not needed), hot-plug I2t 0.091 A2s = 16 % of melting

SURGE ABSORBER / INPUT FILTER



Film: (30 W case, unchanged) 2 x film = RC surge absorber with 44R: 4 kV / 20 us residual -> 1199 V (UR 1300 V); filter corner 546 Hz.

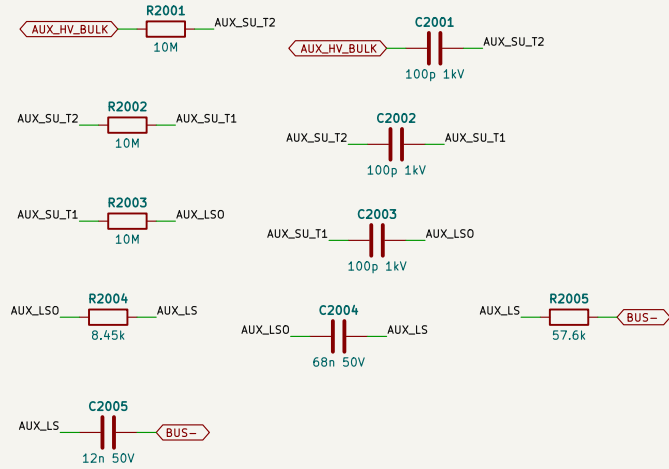
Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 20
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /17 AUX75 input taps, OR, film/
File: 17_aux75_input.kicad_sch

Title: PV-PWR power board – AUX75 input taps, OR, film

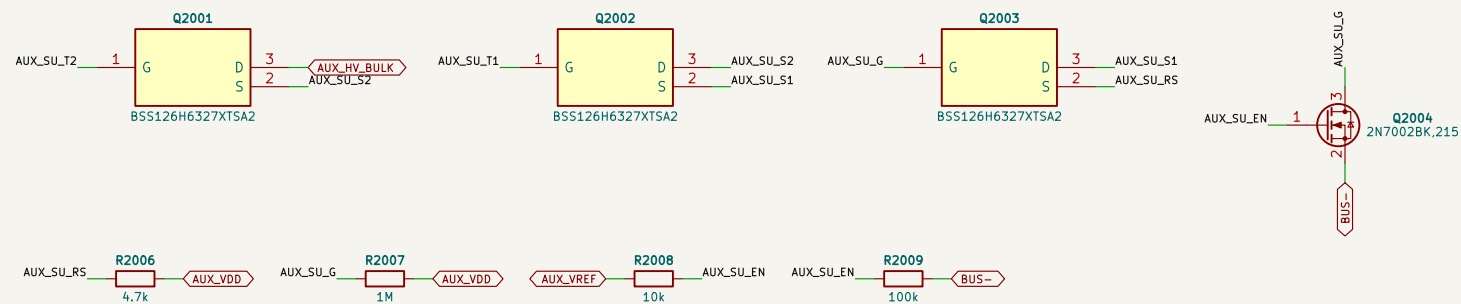
Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 20/23

COMPENSATED TAP STRING



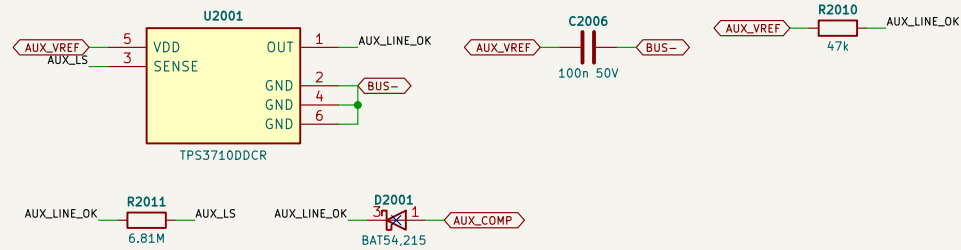
string tap, brown-in 206–215 V, brown-out 179–192 V
brown-in hysteresis (same solve); OV lockout 1107–1162 V

DEPLETION CASCODE CURRENT SOURCE



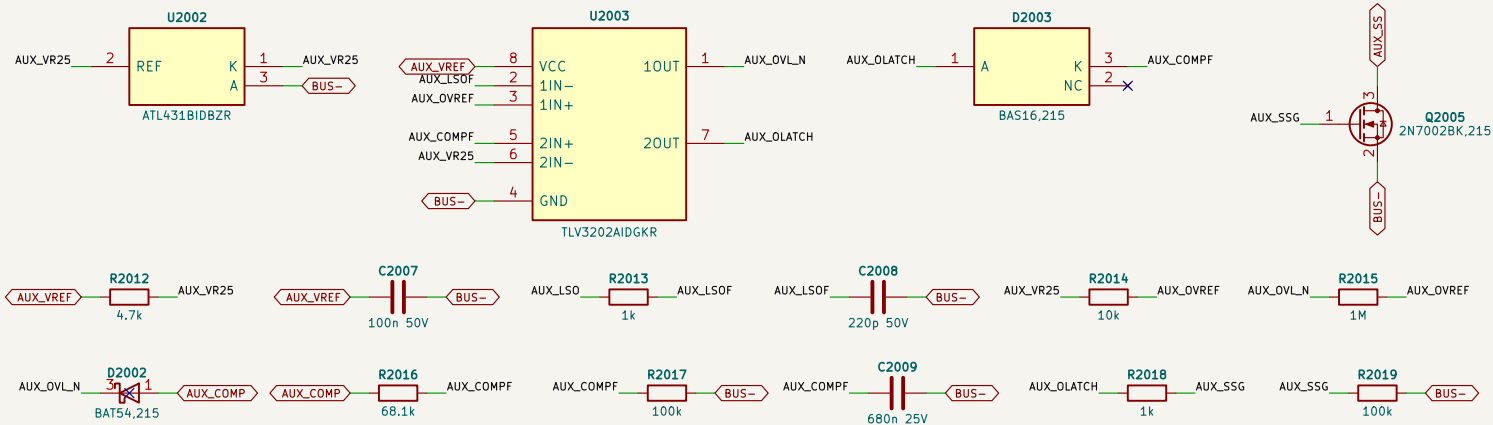
VREF high -> start-up off (as AUX-HV).

LINE BROWN-IN



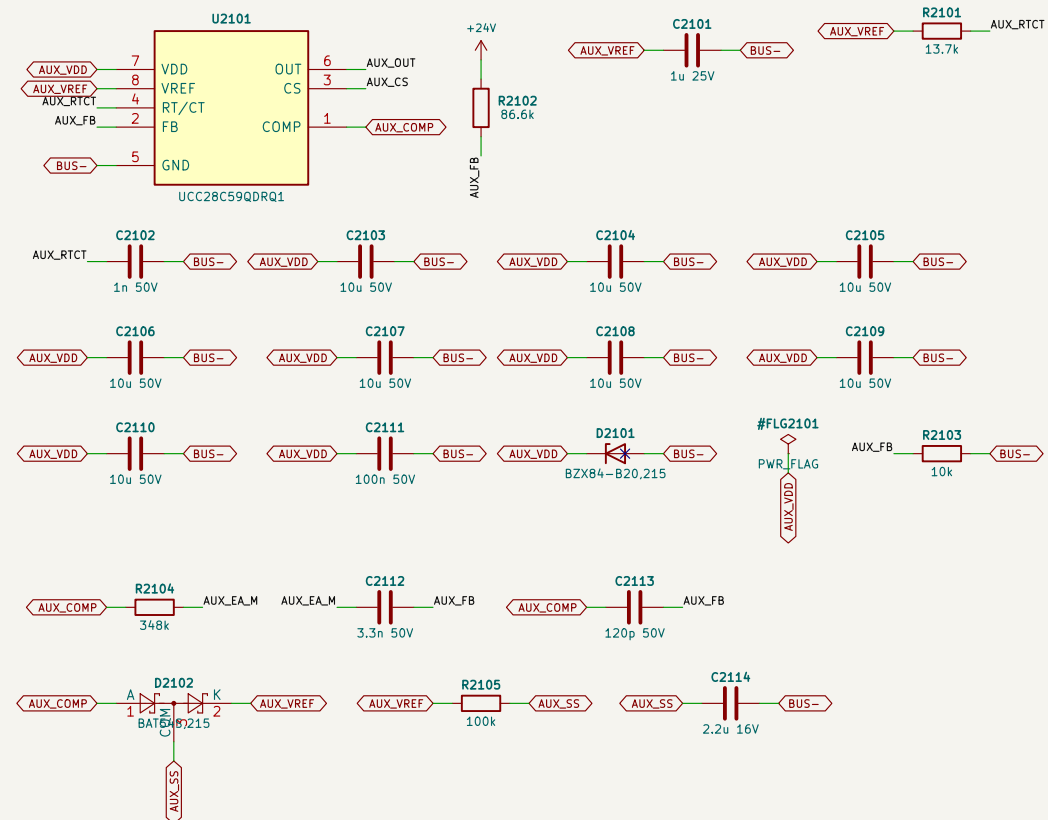
LINE_OK low -> BAT54 holds COMP below the 1.15 V offset.

INPUT OV LOCKOUT AND OVERLOAD TIMER



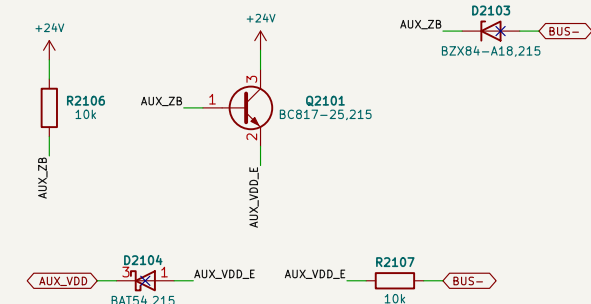
Ch 1: OV lockout -> COMP low. Ch 2: timer divider: trips above 103 / 160 / 256 W (no trip at 75 W + 6 W pull-in); timer 35–80 ms -> latch -> SS crowbar -> hiccup.

PWM CONTROLLER AND ERROR AMPLIFIER



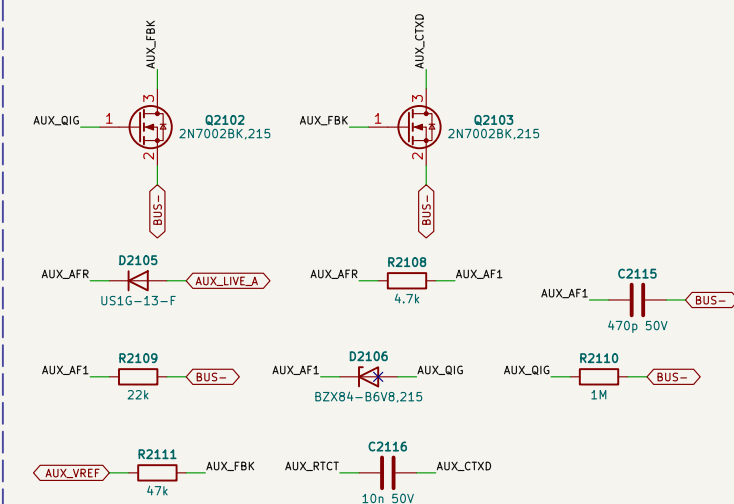
live 24 V into FB: 23.62 / 24.15 / 24.68 V (VFB 2.45–2.55 V, 0.1 %)
type II: mid-band gain 4.0; crossover 241–667 Hz, phase margin ≥ 58 deg type II zero 139 Hz type II pole 3811 Hz

VDD FOLLOWER FROM THE LIVE 24 V



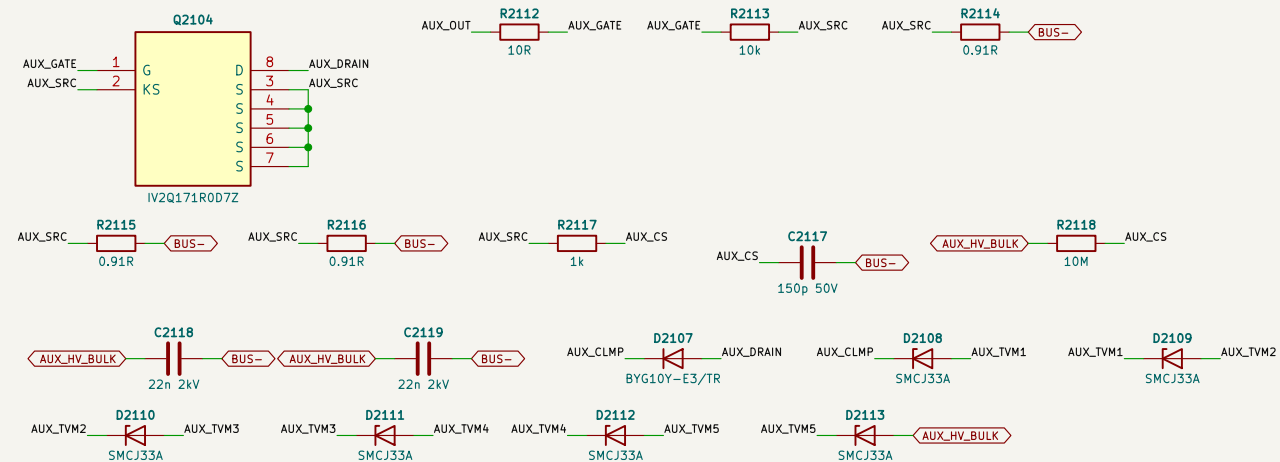
(30 W case, unchanged) follower Zener bias: $V_{DD} = \text{gate drive } 15.46\text{--}18.11 \text{ V}$ (BZX84-A18, $-40..85 \text{ }^{\circ}\text{C}$)

FREQUENCY FOLDBACK ON THE LIVE WINDING PLATEAU (AX-03)



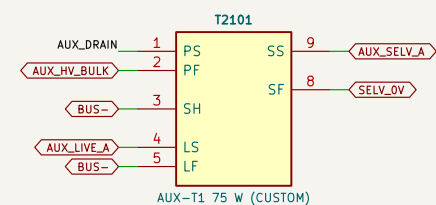
(30 W case, unchanged) foldback: switched onto RT/CT below 7.7–10.1 V output → fsw 5.9 kHz (5.3–6.8)

SIC SWITCH, CURRENT SENSE, LINE FEED-FORWARD, CLAMP



3 In parallel = 0.303333R; lowest limit 2.90 A delivers 103 W ($\geq 110\%$ of 75 W) at every input; highest 3.91 A with the line feed-forward (4.27 A without) line feed-forward HV_BULK $\rightarrow$ CS (CRH2512); CS offset 101 mV at 1000 V cancels most of the 155–250 nS junction overshoot, so the highest current limit no longer rises with the input Clamp: BYG10Y + 6 x SMCJ34; V_DS 1316 / 1420 V at 1000 / 1100 V.

AUX-T1 (CUSTOM – THE MODULE’S REINFORCED SUPPLY BARRIER B1)



Lp 0.485 mH +/-7 %; shield and live winding on BUS-; SELV winding reinforced (PD <= 10 pC routine test).
Electrical requirement: aux75_spec.json transformer_requirement.

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 22
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /19 AUX75 PWM, SiC, clamp, AUX-T1/
File: 19_aux75_power.kicad_sch

Title: PV-PWR power board – AUX75 PWM, SiC, clamp, AUX-T1

Size: A3

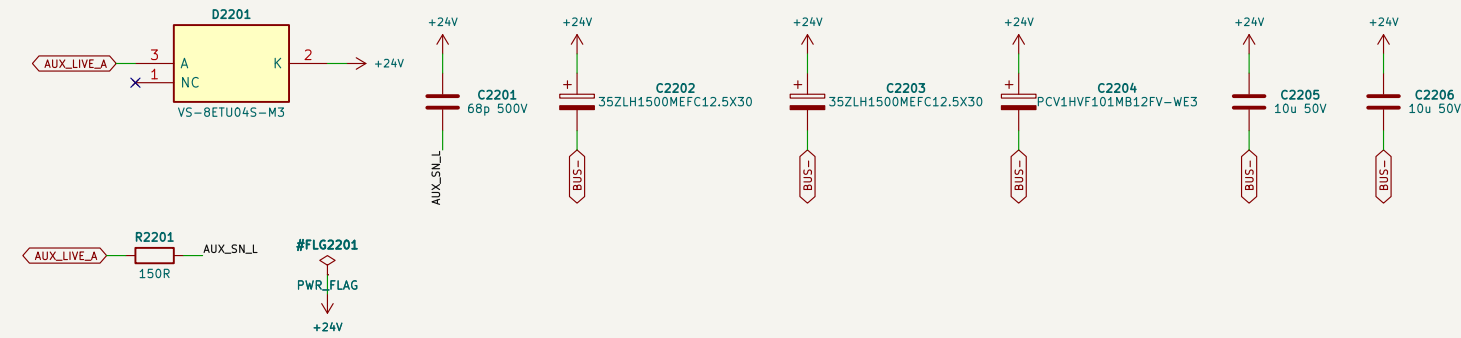
Date: 2026-10-05

Rev: A2

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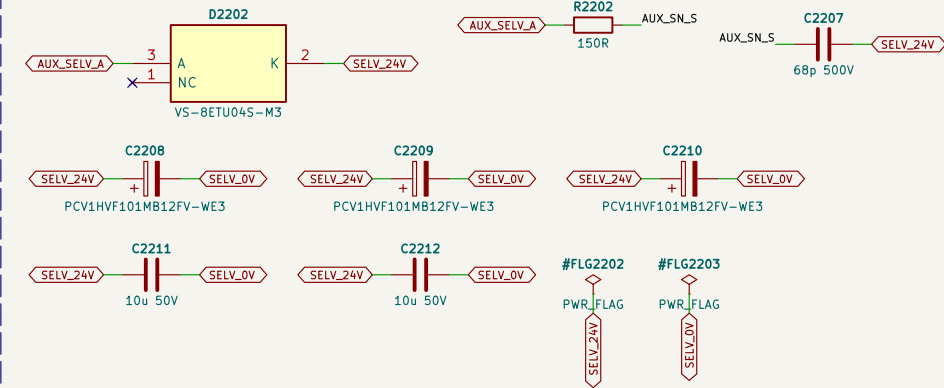
22/23

LIVE 24 V RECTIFIER, HOLD-UP



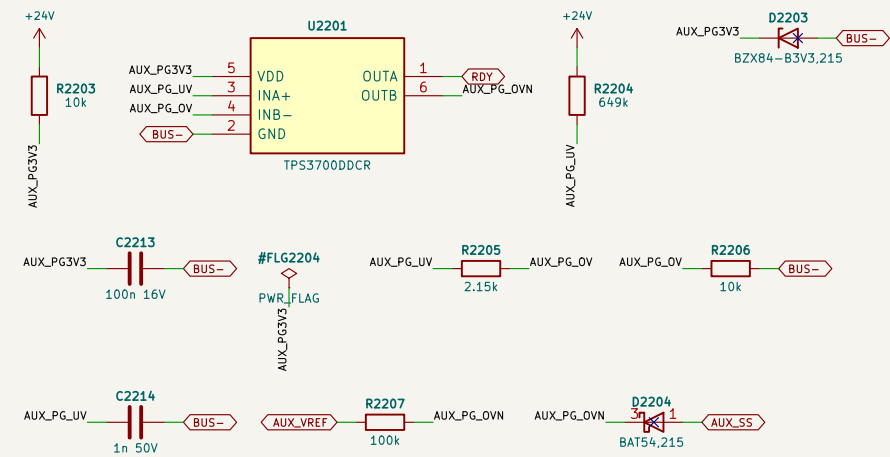
D2PAK tab = cathode on the output. Snubber 150R + 68p.

SELV 24 V RECTIFIER (REINFORCED SIDE)



Cross-regulated from the live winding.

STATUS AND OUTPUT OV STOP (TPS3700 ON A 3.3 V ZENER RAIL)



TPS3700 window on the live 24 V: UV falls 21.02–21.81 V (status), OV rises 26.13–26.76 V (SS pulled low)

Not bench-validated. Values read from sim/out/*.json; checks: design_check()
Sheet 23
Schematics authored by Chinmoy Bhuyan
Live board referenced to BUS-; gate islands functional; SELV winding reinforced; HV-PE basic

Sheet: /20 AUX75 live 24 V, SELV 24 V, status/
File: 20_aux75_outputs.kicad_sch

Title: PV-PWR power board – AUX75 live 24 V, SELV 24 V, status

Size: A3	Date: 2026-10-05	Rev: A2
KiCad E.D.A. 10.0.6		Id: 23/23